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1 Build and Snippet

1.1 Sublime Build

```
{
"cmd" : ["g++ -std=c++20 -DLOCAL -Wall $file_name -o
$file_base_name && timeout 5s ./$file_base_name<input.
txt> output.txt && rm $file_base_name"],
"selector" : "source.cpp",
"shell": true,
"working_dir" : "$file_path"
}
```

2 Data Structures

2.1 BIT

```
class BIT {
  int *bin, N;
  public:
  BIT(int n) : N(n + 1) {
    bin = new int[N + 1];
    memset(bin, 0, (N + 1) * sizeof(int));
  }
  void update(int id, int val) {
    id++;
    for (; id <= N; id += id & -id) bin[id] += val;
  }
  int helper(int id) {
    id++;
    int sum = 0;
    for (; id > 0; id -= id & -id) sum += bin[id];
    return sum;
  }
  int query(int l, int r) { return helper(r) - helper(l
  - 1); }
  ~BIT() { delete[] bin; }
};
```

2.2 Beats

```
class SegTreeBeats { const int INF = INT_MAX; const LL
NEG_INF = LLONG_MIN; vector<LL> mx, mn, smx, smn, sum,
add; vector<int> mxcnt, mncnt; int L, R;
void applyMax(int u, LL x) { sum[u] += mncnt[u] * (x -
mn[u]); if (mx[u] == mn[u]) mx[u] = x; if (smx[u] == mn[
u]) smx[u] = x; mn[u] = x; }
void applyMin(int u, LL x) { sum[u] -= mxcnt[u] * (mx[u]
 - x); if (mn[u] == mx[u]) mn[u] = x; if (smn[u] == mx[u
]) smn[u] = x; mx[u] = x; }
void applyAdd(int u, LL x, int tl, int tr) { sum[u] += (
tr - tl + 1) * x; add[u] += x; mx[u] += x, mn[u] += x;
if (smx[u] != NEG_INF) smx[u] += x; if (smn[u] != INF)
smn[u] += x; }
void push(int u, int tl, int tr) { int lft = u << 1, ryt
 = lft | 1, mid = (tl + tr) >> 1; if (add[u] != 0) {
applyAdd(lft, add[u], tl, mid); applyAdd(ryt, add[u],
mid + 1, tr); add[u] = 0; } if (mx[u] < mx[lft])
applyMin(lft, mx[u]); if (mx[u] < mx[ryt]) applyMin(ryt,
 mx[u]); if (mn[u] > mn[lft]) applyMax(lft, mn[u]); if (
mn[u] > mn[ryt]) applyMax(ryt, mn[u]); }
void merge(int u) { int lft = u << 1, ryt = lft | 1; sum
[u] = sum[lft] + sum[ryt]; mx[u] = max(mx[lft], mx[ryt])
; smx[u] = max(smx[lft], smx[ryt]); if (mx[lft] != mx[
ryt]) smx[u] = max(smx[u], min(mx[lft], mx[ryt])); mxcnt
[u] = (mx[u] == mx[lft]) * mxcnt[lft] + (mx[u] == mx[ryt
]) * mxcnt[ryt]; mn[u] = min(mn[lft], mn[ryt]); smn[u] =
 min(smn[lft], smn[ryt]); if (mn[lft] != mn[ryt]) smn[u]
 = min(smn[u], max(mn[lft], mn[ryt])); mncnt[u] = (mn[u]
 == mn[lft]) * mncnt[lft] + (mn[u] == mn[ryt]) * mncnt[
ryt]; }
void build(const vector<int>& a, int tl, int tr, int u)
{ if (tl == tr) { sum[u] = mn[u] = mx[u] = a[tl]; mxcnt[
u] = mncnt[u] = 1; smx[u] = NEG_INF; smn[u] = INF;
return; } int mid = (tl + tr) >> 1, lft = u << 1, ryt =
lft | 1; build(a, tl, mid, lft); build(a, mid + 1, tr,
ryt); merge(u); }
public:
SegTreeBeats(const vector<int>& a) { int n = a.size(); L
 = 0; R = n - 1; mx.resize(4 * n, 0); mn.resize(4 * n,
0); smx.resize(4 * n, NEG_INF); smn.resize(4 * n, INF);
sum.resize(4 * n, 0); add.resize(4 * n, 0); mxcnt.resize
(4 * n, 0); mncnt.resize(4 * n, 0); build(a, L, R, 1); }
void minimize(int l, int r, LL x) { minimize(l, r, x, L,
 R, 1); } // a[i] = min(x, a[i]);
void maximize(int l, int r, LL x) { maximize(l, r, x, L,
 R, 1); } // a[i] = max(x, a[i]);
void increase(int l, int r, LL x) { increase(l, r, x, L,
 R, 1); } // a[i] = a[i] + x;
LL getSum(int l, int r) { return getSum(l, r, L, R, 1);
}
LL getMax(int l, int r) { return getMax(l, r, L, R, 1);
}
private:
void minimize(int l, int r, LL x, int tl, int tr, int u)
 { if (l > tr || tl > r || mx[u] <= x) return; if (l <=
tl && tr <= r && smx[u] < x) { applyMin(u, x); return; }
 push(u, tl, tr); int mid = (tl + tr) >> 1, lft = u <<
1, ryt = lft | 1; minimize(l, r, x, tl, mid, lft);
minimize(l, r, x, mid + 1, tr, ryt); merge(u); }
void maximize(int l, int r, LL x, int tl, int tr, int u)
 { if (l > tr || tl > r || mn[u] >= x) return; if (l <=
tl && tr <= r && smn[u] > x) { applyMax(u, x); return; }
 push(u, tl, tr); int mid = (tl + tr) >> 1, lft = u <<
1, ryt = lft | 1; maximize(l, r, x, tl, mid, lft);
maximize(l, r, x, mid + 1, tr, ryt); merge(u); }
void increase(int l, int r, LL x, int tl, int tr, int u)
 { if (l > tr || tl > r) return; if (l <= tl && tr <= r)
 { applyAdd(u, x, tl, tr); return; } push(u, tl, tr);
int mid = (tl + tr) >> 1, lft = u << 1, ryt = lft | 1;
increase(l, r, x, tl, mid, lft); increase(l, r, x, mid +
 1, tr, ryt); merge(u); }
LL getSum(int l, int r, int tl, int tr, int u) { if (l >
 tr || tl > r) return 0; if (l <= tl && tr <= r) return
sum[u]; push(u, tl, tr); int mid = (tl + tr) >> 1, lft =
 u << 1, ryt = lft | 1; return getSum(l, r, tl, mid, lft
) + getSum(l, r, mid + 1, tr, ryt); }
LL getMax(int l, int r, int tl, int tr, int u) { if (l >
 tr || tl > r) return NEG_INF; if (l <= tl && tr <= r)
return mx[u]; push(u, tl, tr); int mid = (tl + tr) >> 1;
 return max(getMax(l, r, tl, mid, u << 1), getMax(l, r,
mid + 1, tr, (u << 1) | 1)); } };
```

2.3 Lazy Propagation

```
template <typename T, typename U>
struct LazySegmentTree {
  vector<T> tree;
  vector<U> lazy;
  int n;
  T I;
  function<T(T, T)> merge;
  void updateLazy(int id, U val) { lazy[id] += val; }
  void push(int id, int le, int ri) {
    tree[id] += (ri - le + 1) * lazy[id];
    if (le != ri) {
      updateLazy(2 * id + 1, lazy[id]);
      updateLazy(2 * id + 2, lazy[id]);
    }
    lazy[id] = 0;
  }
  void fix(int id) { tree[id] = merge(tree[2 * id + 1],
  tree[2 * id + 2]); }
```

```
  void build(int id, int le, int ri, vector<T>& arr) {
    if (le == ri) {
      tree[id] = arr[le];
      return;
    }
    int mid = (le + ri) / 2;
    build(2 * id + 1, le, mid, arr);
    build(2 * id + 2, mid + 1, ri, arr);
    fix(id);
  }
  void update(int id, int le, int ri, int l, int r, U
  val) {
    push(id, le, ri);
    if (r < le || l > ri) return;
    if (le >= l && ri <= r) {
      updateLazy(id, val), push(id, le, ri);
      return;
    }
    int mid = (le + ri) / 2;
    update(2 * id + 1, le, mid, l, r, val);
    update(2 * id + 2, mid + 1, ri, l, r, val);
    fix(id);
  }
  T query(int id, int le, int ri, int l, int r) {
    push(id, le, ri);
    if (r < le || l > ri) return I;
    if (l <= le && ri <= r) return tree[id];
    int mid = (le + ri) / 2;
    return merge(query(2 * id + 1, le, mid, l, r),
                 query(2 * id + 2, mid + 1, ri, l, r));
  }
  int findL(int id, int le, int ri, int l, int r, T val)
   {
    push(id, le, ri);
    if (r < le || l > ri) return r + 1;
    if (l <= le && ri <= r && tree[id] > val) return r +
     1;
    if (le == ri) return le;

    int mid = (le + ri) / 2;
    int left = findL(2 * id + 1, le, mid, l, r, val);
    if (left <= r) return left;
    return findL(2 * id + 2, mid + 1, ri, l, r, val);
  }
public:
  LazySegmentTree(vector<T>& arr, T I, function<T(T, T)>
   merge)
      : n(arr.size()), I(I), merge(merge) {
    tree.assign(4 * n, I);
    lazy.assign(4 * n, 0);
    build(0, 0, n - 1, arr);
  }
  LazySegmentTree(int n, T I, function<T(T, T)> merge)
      : n(n), I(I), merge(merge) {
    tree.assign(4 * n, I);
    lazy.assign(4 * n, 0);
  }
  void update(int l, int r, U val) { update(0, 0, n - 1,
   l, r, val); }
  T query(int l, int r) { return query(0, 0, n - 1, l, r
  ); }
  int findL(int l, int r, T val) { return findL(0, 0, n
  - 1, l, r, val); }
};
```

2.4 MO With Update

```
const int N = 1e5 + 5, sz = 2700, bs = 25; int arr[N],
freq[2 * N], cnt[2 * N], id[N], ans[N]; struct query {
int l, r, t, L, R; query(int l = 1, int r = 0, int t =
1, int id = -1) : l(l), r(r), t(t), L(l / sz), R(r / sz)
 {} bool operator<(const query& rhs) const { return (L <
 rhs.L) or (L == rhs.L and R < rhs.R) or (L == rhs.L and
 R == rhs.R and t < rhs.t); } } Q[N]; struct update {
int idx, val, last; } Up[N]; int qi = 0, ui = 0; int l =
 1, r = 0, t = 0; void add(int idx) { --cnt[freq[arr[idx
]]]; freq[arr[idx]]++; cnt[freq[arr[idx]]]++; } void
remove(int idx) { --cnt[freq[arr[idx]]]; freq[arr[idx
]]--; cnt[freq[arr[idx]]]++; } void apply(int t) { const
 bool f = l <= Up[t].idx and Up[t].idx <= r; if (f)
remove(Up[t].idx); arr[Up[t].idx] = Up[t].val; if (f)
add(Up[t].idx); } void undo(int t) { const bool f = l <=
 Up[t].idx and Up[t].idx <= r; if (f) remove(Up[t].idx);
 arr[Up[t].idx] = Up[t].last; if (f) add(Up[t].idx); }
int mex() { for (int i = 1; i <= N; i++) if (!cnt[i])
return i; assert(0); } int main() { int n, q; cin >> n
>> q; int counter = 0; map<int, int> M; for (int i = 1;
i <= n; i++) { cin >> arr[i]; if (!M[arr[i]]) M[arr[i]]
= ++counter; arr[i] = M[arr[i]]; } iota(id, id + N, 0);
while (q--) { int tp, x, y; cin >> tp >> x >> y; if (tp
== 1) Q[++qi] = query(x, y, ui); else { if (!M[y]) M[y]
= ++counter; y = M[y]; Up[++ui] = {x, y, arr[x]}; arr[x]
 = y; } } t = ui; cnt[0] = 3 * n; sort(id + 1, id + qi +
 1, [&](int x, int y) { return Q[x] < Q[y]; }); for (int
 i = 1; i <= qi; i++) { int x = id[i]; while (Q[x].t > t
) apply(++t); while (Q[x].t < t) undo(t--); while (Q[x].
l < l) add(--l); while (Q[x].r > r) add(++r); while (Q[x
].l > l) remove(l++); while (Q[x].r < r) remove(r--);
ans[x] = mex(); } for (int i = 1; i <= qi; i++) cout <<
ans[i] << '\n'; }
```

2.5 MO

```
struct node { LL l, r, idx; };
bool cmp(const node &x, const node &y) {
  return x.r < y.r;
}
LL curr = 0, BLOCK_SIZE, m;
vector<node> query[M];
vector<LL> mp(N, 0);
void add(LL x) {
  if(!mp[x]) curr++;
  mp[x]++;
}
void diminish(LL x) {
  mp[x]--;
  if(!mp[x]) curr--;
}
void solve() {
  int n, q; cin >> n >> q;
  vector<int> v(n), answer(q);
  for(int &i : v) cin >> i;
  BLOCK_SIZE = sqrt(n) + 1, m = 0;
  for(int i=0;i<q;i++) {
    LL x, y; cin >> x >> y; x--; y--;
    query[x / BLOCK_SIZE].push_back({x, y, i});
    m = max(m, x / BLOCK_SIZE);
  }
  for(int i=0;i<=m;i++) sort(all(query[i]), cmp);
  LL mo_left = 0, mo_right = -1;
  for(int i=0;i<=m;i++) {
    for(auto cur : query[i]) {
      LL left = cur.l, right = cur.r, id = cur.idx;
      while(mo_right < right) add(v[++mo_right]);
      while(mo_right > right) diminish(v[mo_right--]);
      while(mo_left < left) diminish(v[mo_left++]);
      while(mo_left > left) add(v[--mo_left]);
      answer[id] = curr;
    }
```

```
  }
  for(int i=0;i<q;i++) cout << answer[i] << '\n';
}
```

2.6 PST

```
// this calculates xor/xor_hash of all the element less
than 'x' in [0, i].
// query is a walk function
class PST {
#define lc(u) tree[u].left
#define rc(u) tree[u].right;
  struct node {
    int left = 0, right = 0, val = 0;
  };
  node* tree;
  int N, LG, time = 0, I = 0;

  node create(int l, int r) { return {l, r, merge(tree[l
  ].val, tree[r].val)}; }
  int merge(LL a, LL b) { return a ^ b; }
  int build(int le, int ri) {
    int id = ++time;
    if (le == ri) return tree[id] = node(), id;
    int m = (le + ri) / 2;
    return tree[id] = create(build(le, m), build(m + 1,
    ri)), id;
  }
  int update(int id, int le, int ri, int pos, int val) {
    int nid = ++time;
    if (le == ri)
      return tree[nid] = {0, 0, (tree[id].val ^ val)},
      nid; // change here
    int m = (le + ri) / 2;
    if (pos <= m) {
      tree[nid] =
          create(update(tree[id].left, le, m, pos, val),
          tree[id].right);
    } else {
      tree[nid] =
          create(tree[id].left, update(tree[id].right, m
          + 1, ri, pos, val));
    }
    return nid;
  }
  int query(int id, int di, int le, int ri) {
    if (tree[id].val == tree[di].val) return 0;
    if (le == ri) return le;
    int m = (le + ri) >> 1;
    if (tree[tree[id].left].val != tree[tree[di].left].
    val)
      return query(tree[id].left, tree[di].left, le, m);
    return query(tree[id].right, tree[di].right, m + 1,
    ri);
  }

 public:
  PST(int N, int U) { // U --> number of expected
  updates
    this->N = N;
    LG = 33 - __builtin_clz(N);
    tree = new node[N * 4 + U * LG];
    build(0, N - 1);
  }
  int update(int id, int pos, int val) {
    return update(id, 0, N - 1, pos, val);
  }
  int query(int id, int di) { return query(id, di, 0, N
  - 1); }
  ~PST() { delete[] tree; }
};
```

2.7 Sparse Table

```
class SparsedTable {
 private:
  vector<vector<int>> table;
  vector<int> log;
  int n;
 public:
  SparsedTable(const vector<int>& arr) {
    n = arr.size(); log.resize(n + 1); buildLog();
    table = vector<vector<int>>(log[n] + 1, vector<int>(
    n));
    for (int i = 0; i < n; i++) {
      table[0][i] = arr[i];
    }
    for (int j = 1; j <= log[n]; j++) {
      for (int i = 0; i + (1 << j) <= n; i++) {
        table[j][i] = merge(table[j - 1][i], table[j -
        1][i + (1 << (j - 1))]);
      }}}
  int merge(int a, int b) { return a | b; }
  void buildLog() {
    log[1] = 0;
    for (int i = 2; i <= n; i++) {
      log[i] = log[i / 2] + 1;
    }}
  int Query(int L, int R) {
    int j = log[R - L + 1];
    return merge(table[j][L], table[j][R - (1 << j) +
    1]);
  }
  void update(int L, int R, int val) {
    int j = log[R - L + 1];
    table[j][L] = min(table[j][L], val);
    table[j][R - (1 << j) + 1] = min(table[j][R - (1 <<
    j) + 1], val);
  }
  void finalize() {
    for (int j = log[n]; j > 0; j--) {
      for (int i = 0; i + (1 << j) <= n; i++) {
        table[j - 1][i] = min(table[j - 1][i], table[j][i
        ]);
        table[j - 1][i + (1 << (j - 1))] =
            min(table[j - 1][i + (1 << (j - 1))], table[j
            ][i]);
      }}}};
```

2.8 Trie

```
struct Trie {
  static const int B = 31;
  struct node {
    node* nxt[2];
    int sz;
    node() {
      nxt[0] = nxt[1] = NULL;
      sz = 0;
    }
  }*root;
  Trie() {root = new node();}
  void insert(int val) {
    node* cur = root;
    cur -> sz++;
    for (int i = B - 1; i >= 0; i--) {
      int b = val >> i & 1;
      if (cur -> nxt[b] == NULL) cur -> nxt[b] = new node
      ();
      cur = cur -> nxt[b];
      cur -> sz++;
```

```
  }}
int query(int x, int k) { // number of values s.t. val
 ^ x < k
  node* cur = root;
  int ans = 0;
  for (int i = B - 1; i >= 0; i--) {
    if (cur == NULL) break;
    int b1 = x >> i & 1, b2 = k >> i & 1;
    if (b2 == 1) {
      if (cur -> nxt[b1]) ans += cur -> nxt[b1] -> sz;
      cur = cur -> nxt[!b1];
    } else cur = cur -> nxt[b1];
  }
  return ans;
}
int get_max(int x) { // returns maximum of val ^ x
  node* cur = root;
  int ans = 0;
  for (int i = B - 1; i >= 0; i--) {
    int k = x >> i & 1;
    if (cur -> nxt[!k]) cur = cur -> nxt[!k], ans <<=
    1, ans++;
    else cur = cur -> nxt[k], ans <<= 1;
  }
  return ans;
}
int get_min(int x) { // returns minimum of val ^ x
  node* cur = root;
  int ans = 0;
  for (int i = B - 1; i >= 0; i--) {
    int k = x >> i & 1;
    if (cur -> nxt[k]) cur = cur -> nxt[k], ans <<= 1;
    else cur = cur -> nxt[!k], ans <<= 1, ans++;
  }
  return ans;
}
void del(node* cur) {
  for (int i = 0; i < 2; i++) if (cur -> nxt[i]) del(
  cur -> nxt[i]);
  delete(cur);
}} t;
```

2.9 segtree iterative

```
template <class T, class L, T it, T (*mergeT)(T, T), T
(*applyL)(T, L)> struct SegTree { int n; vector<T> tree;
 void update(int pos, L val) { pos += n; tree[pos] =
applyL(tree[pos], val); while (pos > 1) { pos >>= 1;
tree[pos] = mergeT(tree[pos << 1], tree[pos << 1 | 1]);
} } T query(int l, int r) { l += n, r += n + 1; T lft =
it, ryt = it; while (l < r) { if (l & 1) lft = mergeT(
lft, tree[l++]); if (r & 1) ryt = mergeT(tree[--r], ryt)
; l >>= 1; r >>= 1; } return mergeT(lft, ryt); } SegTree
(vector<T>& v) : n(v.size()) { tree.assign(2 * n, it);
for (int i = 0; i < n; i++) tree[i + n] = v[i]; for (int
 i = n - 1; i > 0; i--) tree[i] = mergeT(tree[i << 1],
tree[i << 1 | 1]); } };
```

2.10 sparse table 2D

```
// rectangle query
namespace st2 {
  const int N = 2e3 + 5, B = 12;
  using Ti = long long;
  Ti Id = LLONG_MAX;
  Ti f(Ti a, Ti b) { return max(a, b); }
  Ti tbl[N][N][B];
  void init(int n, int m) {
    for(int k = 1; k < B; k++) {
     for(int i = 0; i + (1 << k) - 1 < n; i++) {
      for(int j = 0; j + (1 << k) - 1 < m; j++) {
       tbl[i][j][k] = tbl[i][j][k - 1];
       tbl[i][j][k] = f(tbl[i][j][k], tbl[i][j + (1 << k
       - 1)][k - 1]);
       tbl[i][j][k] = f(tbl[i][j][k], tbl[i + (1 << k -
       1)][j][k - 1]);
       tbl[i][j][k] = f(tbl[i][j][k], tbl[i + (1 << k -
       1)][j + (1 << k - 1)][k - 1]);
    } } }
  }
  Ti query(int i, int j, int len) {
    int k = __lg(len);
    LL ret = tbl[i][j][k];
    ret = f(ret, tbl[i + len - (1 << k)][j][k]);
    ret = f(ret, tbl[i][j + len - (1 << k)][k]);
    ret = f(ret, tbl[i + len - (1 << k)][j + len - (1 <<
     k)][k]);
    return ret;
  }
}
int main() {
  for(int i = 0; i < n; i++)
    for(int j = 0; j < m; j++)
      cin >> st2 :: tbl[i][j][0];
  st2 :: init(n, m);
  cout << st2 :: query(x, y, s); // x, y, x + s - 1, y +
   s - 1
}
```

3 Number Theory

3.1 Chinese Reminder Theorem

```
// given a, b will find solutions for, ax + by = 1
tuple<LL, LL, LL> EGCD(LL a, LL b) {
  if (b == 0) return {1, 0, a};
  else {
    auto [x, y, g] = EGCD(b, a % b);
    return {y, x - a / b * y, g};
  }
} // given modulo equations, will apply CRT
PLL CRT(vector<PLL>& v) {
  LL V = 0, M = 1;
  for (auto& [v, m] : v) { // value % mod
    auto [x, y, g] = EGCD(M, m);
    if ((v - V) % g != 0) return {-1, 0};
    V += x * (v - V) / g % (m / g) * M, M *= m / g;
    V = (V % M + M) % M;
  }
  return make_pair(V, M);
}
```

3.2 FFT

```
// typedef complex<double> CD;
struct CD {
  double x, y;
  CD(double x = 0, double y = 0) : x(x), y(y) {}
  CD operator+(const CD &o) { return {x + o.x, y + o.y};
   }
  CD operator-(const CD &o) { return {x - o.x, y - o.y};
   }
  CD operator*(const CD &o) { return {x * o.x - y * o.y,
   x * o.y + o.x * y}; }
  void operator/=(double d) { x /= d; y /= d; }
  double real() { return x; }
  double imag() { return y; }
};
CD conj(const CD &c) { return CD(c.x, -c.y); }
typedef long long LL;
const double PI = acos(-1.0L);
namespace FFT {
```

```
int N; vector<int> perm;
vector<CD> wp[2];
void precalculate(int n) {
  assert((n & (n - 1)) == 0);
  N = n; perm = vector<int>(N, 0);
  for (int k = 1; k < N; k <<= 1) {
    for (int i = 0; i < k; i++) {
      perm[i] <<= 1;
      perm[i + k] = 1 + perm[i];
  } }
  wp[0] = wp[1] = vector<CD>(N);
  for (int i = 0; i < N; i++) {
    wp[0][i] = CD(cos(2 * PI * i / N), sin(2 * PI * i /
    N));
    wp[1][i] = CD(cos(2 * PI * i / N), -sin(2 * PI * i /
     N));
} }
void fft(vector<CD> &v, bool invert = false) {
  if (v.size() != perm.size()) precalculate(v.size());
  for (int i = 0; i < N; i++)
    if (i < perm[i]) swap(v[i], v[perm[i]]);
  for (int len = 2; len <= N; len *= 2) {
    for (int i = 0, d = N / len; i < N; i += len) {
      for (int j = 0, idx = 0; j < len / 2; j++, idx += d
      ) {
        CD x = v[i + j];
        CD y = wp[invert][idx] * v[i + j + len / 2];
        v[i + j] = x + y;
        v[i + j + len / 2] = x - y;
  } } }
  if (invert) { for (int i = 0; i < N; i++) v[i] /= N; }
}
void pairfft(vector<CD> &a, vector<CD> &b, bool invert =
 false) {
  int N = a.size();
  vector<CD> p(N);
  for (int i = 0; i < N; i++) p[i] = a[i] + b[i] * CD(0,
   1);
  fft(p, invert);
  p.push_back(p[0]);
  for (int i = 0; i < N; i++) {
    if (invert) {
      a[i] = CD(p[i].real(), 0);
      b[i] = CD(p[i].imag(), 0);
    } else {
      a[i] = (p[i] + conj(p[N - i])) * CD(0.5, 0);
      b[i] = (p[i] - conj(p[N - i])) * CD(0, -0.5);
} } }
vector<bool> multiply(const vector<bool> &a, const
vector<bool> &b) {
  int n = 1;
  while (n < a.size() + b.size()) n <<= 1;
  vector<CD> fa(a.begin(), a.end()), fb(b.begin(), b.end
  ());
  fa.resize(n);
  fb.resize(n);
  //       fft(fa); fft(fb);
  pairfft(fa, fb);
  for (int i = 0; i < n; i++) fa[i] = fa[i] * fb[i];
  fft(fa, true);
  vector<bool> ans(n);
  for (int i = 0; i < n; i++) ans[i] = round(fa[i].real
  ());
  while (ans.size() > 1 && !ans.back()) ans.pop_back();
  return ans;
}
const int M = 1e9 + 7, B = sqrt(M) + 1;
vector<LL> anyMod(const vector<LL> &a, const vector<LL>
&b) {
  int n = 1;
  while (n < a.size() + b.size()) n <<= 1;
  vector<CD> al(n), ar(n), bl(n), br(n);
  for (int i = 0; i < a.size(); i++) al[i] = a[i] % M /
  B, ar[i] = a[i] % M % B;
  for (int i = 0; i < b.size(); i++) bl[i] = b[i] % M /
  B, br[i] = b[i] % M % B;
  pairfft(al, ar);
  pairfft(bl, br);
  //       fft(al); fft(ar); fft(bl); fft(br);
  for (int i = 0; i < n; i++) {
    CD ll = (al[i] * bl[i]), lr = (al[i] * br[i]);
    CD rl = (ar[i] * bl[i]), rr = (ar[i] * br[i]);
    al[i] = ll; ar[i] = lr;
    bl[i] = rl; br[i] = rr;
  }
  pairfft(al, ar, true);
  pairfft(bl, br, true);
  //       fft(al, true); fft(ar, true); fft(bl, true);
  fft(br, true);
  vector<LL> ans(n);
  for (int i = 0; i < n; i++) {
    LL right = round(br[i].real()), left = round(al[i].
    real());
    LL mid = round(round(bl[i].real()) + round(ar[i].
    real()));
    ans[i] = ((left % M) * B * B + (mid % M) * B + right
    ) % M;
  }
  return ans;
}
} // namespace FFT
```

3.3 Matrix

```
template <typename DT>
struct Matrix {
  vector<vector<DT>> mat;
  Matrix(vector<vector<DT>>& mat) : mat(mat) {}
  Matrix(int n) : mat(n, vector<DT>(n)) {
    for (int i = 0; i < n; i++) {
      mat[i][i] = 1;
  } }
  Matrix(int n, int m) : mat(n, vector<DT>(m)) {}
  Matrix operator*(Matrix& other) {
    auto &a = mat, &b = other.mat;
    assert(a[0].size() == b.size());
    int n = a.size(), m = b[0].size(), s = a[0].size();
    Matrix<DT> ret(n, m);
    for (int i = 0; i < n; i++) {
      for (int j = 0; j < m; j++) {
        DT temp = 0;
        for (int k = 0; k < s; k++) {
          temp = (temp + 1LL * a[i][k] * b[k][j] % MOD) %
           MOD;
        }
        ret.mat[i][j] = temp;
    } }
    return ret;
} };
```

3.4 NOD and SOD

```
// NUMBER = p_1^a_1 * p_2^a_2 .... p_n^a_n
LL NOD = 1, SOD = 1, POD = 1, POWER = 1;
for (int i = 0; i < n; i++) {
  LL p, a; cin >> p >> a;
  NOD = (NOD * (a + 1)) % MOD;
  SOD = ((SOD * (bigmod(p, a + 1, MOD) + MOD - 1)) % MOD
   * inv[p - 1]) % MOD;
```

```
  POD = bigmod(POD, a + 1, MOD) * bigmod(bigmod(x, a * (
  a + 1) / 2, MOD), POWER, MOD) % MOD;
  POWER = (POWER * (a + 1)) % (MOD - 1);
}
LL csod(LL n) {
  LL ans = 0;
  for (LL i = 2; i * i <= n; ++i) {
    LL j = n / i;
    ans += (i + j) * (j - i + 1) / 2;
    ans += i * (j - i);
  }
  return ans;
}
// summation of NOD(d)[d | n] = product of g(e_k + 1)[n
= p_k ^ a_k] g(x) = x * (x + 1) / 2
```

3.5 Pollard rho

```
namespace rho { inline LL mul(LL a, LL b, LL mod) { LL
result = 0; while (b) { if (b & 1) result = (result + a)
 % mod; a = (a + a) % mod; b >>= 1; } return result; }
LL bigmod(LL num, LL pow, LL mod){} inline bool is_prime
(LL n) { if (n < 2 or n % 6 % 4 != 1) return (n | 1) ==
3; LL a[] = {2, 325, 9375, 28178, 450775, 9780504,
1795265022}; LL s = __builtin_ctzll(n - 1), d = n >> s;
for (LL x : a) { LL p = bigmod(x % n, d, n), i = s; for
(; p != 1 and p != n - 1 and x % n and i--; p = mul(p, p
, n)); if (p != n - 1 and i != s) return false; } return
 true; } LL f(LL x, LL n) { return mul(x, x, n) + 1; }
LL get_factor(LL n) { LL x = 0, y = 0, t = 0, prod = 2,
i = 2, q; for (; t++ % 40 or __gcd(prod, n) == 1; x = f(
x, n), y = f(f(y, n), n)) { (x == y) ? x = i++, y = f(x,
 n) : 0; prod = (q = mul(prod, max(x, y) - min(x, y), n)
) ? q : prod; } return __gcd(prod, n); } void _factor(LL
 n, map<LL, int>& res) { if (n == 1) return; if (
is_prime(n)) res[n]++; else { LL x = get_factor(n);
_factor(x, res); _factor(n / x, res); } } map<LL, int>
factorize(LL n) { map<LL, int> res; if (n < 2) return
res; LL small_primes[] = {2, 3, 5, 7, 11, 13, 17, 19,
23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79,
83, 89, 97}; for (LL p : small_primes) for (; n % p ==
0; n /= p, res[p]++); _factor(n, res); return res; } }
// namespace rho
```

3.6 Sieve

```
const int N = 10000000;
vector<int> lp(N), pr;
for (int i = 2; i < N; i++) {
  if (lp[i] == 0) {
    lp[i] = i; pr.push_back(i);
  }
  for (int j = 0; i * pr[j] < N; j++) {
    lp[i * pr[j]] = pr[j];
    if (pr[j] == lp[i]) break;
} }
```

3.7 XOR Basis

```
template <typename T = int, int B = 31>
struct Basis {
  T a[B];
  Basis() { memset(a, 0, sizeof a); }
  void insert(T x) {
    for (int i = B - 1; i >= 0; i--) {
      if (x >> i & 1) {
        if (a[i]) x ^= a[i];
        else { a[i] = x; break; }
  } } }
  bool can(T x) {
    for (int i = B - 1; i >= 0; i--)
      x = min(x, x ^ a[i]);
    return x == 0;
  }
  T max_xor(T ans = 0) {
    for (int i = B - 1; i >= 0; i--)
      ans = max(ans, ans ^ a[i]);
    return ans;
} };
```

3.8 gauss

```
template<int MOD> struct Mint { long long val; Mint(long
 long v = 0) { val = v % MOD; if (val < 0) val += MOD; }
 Mint operator + (const Mint &other) const { return Mint
(val + other.val); } Mint operator - (const Mint &other)
 const { return Mint(val - other.val); } Mint operator *
 (const Mint &other) const { return Mint(val * other.val
); } static Mint power(Mint a, long long p) { Mint res =
 1; while (p) { if (p & 1) res = res * a; a = a * a; p
>>= 1; } return res; } Mint inv() const { // MOD must be
 prime return power(*this, MOD - 2); } Mint operator / (
const Mint &other) const { return *this * other.inv(); }
 };
/* ================= MODULAR GAUSSIAN ELIMINATION
================= */
template<typename T>
struct Gauss {
    int n, m;                   // n equations, m variables
    vector<vector<T>> a;        // augmented matrix
    vector<T> solution;
    T determinant = 1;
    Gauss(int n, int m) : n(n), m(m) {
        a.assign(n, vector<T>(m + 1));
    }
    void set_row(int i, const vector<T>& row) {
        a[i] = row;             // size must be m+1
    }
    // returns:
    // 0 = no solution
    // 1 = unique solution
    // 2 = infinite solutions
    int solve() {
        int rank = 0;
        determinant = 1;
        for (int col = 0; col < m && rank < n; col++) {
            int pivot = -1;
            for (int i = rank; i < n; i++) {
                if (a[i][col].val != 0) {
                    pivot = i;
                    break;
                }
            }
            if (pivot == -1) {
                determinant = 0;
                continue;
            }
            swap(a[pivot], a[rank]);
            if (pivot != rank)
                determinant = determinant * T(-1);
            determinant = determinant * a[rank][col];
            T inv_pivot = T(1) / a[rank][col];
            // normalize pivot row
            for (int j = col; j <= m; j++)
                a[rank][j] = a[rank][j] * inv_pivot;
            // eliminate other rows
            for (int i = 0; i < n; i++) {
                if (i != rank && a[i][col].val != 0) {
                    T factor = a[i][col];
                    for (int j = col; j <= m; j++)
                        a[i][j] = a[i][j] - factor * a[
                        rank][j];
```

```
            }
        }
        rank++;
    }
    // check consistency
    for (int i = rank; i < n; i++) {
        if (a[i][m].val != 0)
            return 0; // no solution
    }
    if (rank < m)
        return 2; // infinite solutions
    // unique solution
    solution.assign(m, 0);
    for (int i = 0; i < m; i++)
        solution[i] = a[i][m];

    return 1;
  }
  vector<T> get_solution() const {return solution;}
  T get_determinant() const {return determinant;}
};
```

3.9 linear diophantine

```
using PLL = pair <LL,LL>;
// EGCD
tuple <bool, LL, LL, LL> get_solution(LL a, LL b, LL c)
{
 auto [x, y, g] = EGCD(abs(a), abs(b));
 x *= (a < 0) ? -c / g : c / g;
 y *= (b < 0) ? -c / g : c / g;
 return {c % g == 0, x, y, g} ; }
PLL shift_solution(LL a, LL b, LL x, LL y, LL shift) {
 return {x + shift * b, y - shift * a}; }
```

3.10 mobius function

```
const int N = 1e6 + 5;
int mob[N];
void mobius() {
  memset(mob, -1, sizeof mob);
  mob[1] = 1;
  for (int i = 2; i < N; i++)
    if (mob[i]) {
      for (int j = i + i; j < N; j += i) mob[j] -= mob[i
      ];
} }
```

3.11 nCr

```
namespace com {
LL fact[N], inv[N], inv_fact[N];
void init() {
  fact[0] = inv_fact[0] = 1;
  for (LL i = 1; i < N; i++) {
    inv[i] = i == 1 ? 1 : (LL)inv[i - mod % i] * (mod /
    i + 1) % mod;
    fact[i] = (LL)fact[i - 1] * i % mod;
    inv_fact[i] = (LL)inv_fact[i - 1] * inv[i] % mod;
} }
LL C(int n, int r) {
  return (r < 0 or r > n) ? 0 : fact[n] * inv_fact[r] %
  mod * inv_fact[n - r] % mod;
} } // namespace com
```

3.12 ntt

```
const LL N = 1 << 18;
const LL MOD = 786433;
vector<LL> P[N];
LL rev[N], w[N | 1], a[N], b[N], inv_n, g =
primitive_root (MOD);
// LL Pow = bigmod
LL primitive_root(LL p) {
  vector<LL> factor;
  LL phi = p - 1, n = phi;
  for (LL i = 2; i * i <= n; i++) {
    if (n % i) continue;
    factor.emplace_back(i);
    while (n % i == 0) n /= i;
  }
  if (n > 1) factor.emplace_back(n);
  for (LL res = 2; res <= p; res++) {
    bool ok = true;
    for (LL i = 0; i < factor.size() && ok; i++)
      ok &= Pow(res, phi / factor[i]) != 1;
    if (ok) return res;
  }
  return -1;
}
void prepare(LL n) {
  LL sz = abs(31 - __builtin_clz(n));
  LL r = Pow(g, (MOD - 1) / n);
  inv_n = Pow(n, MOD - 2);
  w[0] = w[n] = 1;
  for (LL i = 1; i < n; i++) w[i] = (w[i - 1] * r) % MOD
  ;
  for (LL i = 1; i < n; i++)
    rev[i] = (rev[i >> 1] >> 1) | ((i & 1) << (sz - 1));
}
void NTT(LL *a, LL n, LL dir = 0) {
  for (LL i = 1; i < n - 1; i++)
    if (i < rev[i]) swap(a[i], a[rev[i]]);
  for (LL m = 2; m <= n; m <<= 1) {
    for (LL i = 0; i < n; i += m) {
      for (LL j = 0; j < (m >> 1); j++) {
        LL &u = a[i + j], &v = a[i + j + (m >> 1)];
        LL t = v * w[dir ? n - n / m * j : n / m * j] %
        MOD;
        v = u - t < 0 ? u - t + MOD : u - t;
        u = u + t >= MOD ? u + t - MOD : u + t;
  } } }
  if (dir) for (LL i = 0; i < n; i++) a[i] = (inv_n * a[
  i]) % MOD;
}
vector<LL> multiply (vector<LL> p, vector<LL> q) {
  LL n = p.size(), m = q.size();
  LL t = n + m - 1, sz = 1;
  while (sz < t) sz <<= 1;
  prepare(sz);
  for (LL i = 0; i < n; i++) a[i] = p[i];
  for (LL i = 0; i < m; i++) b[i] = q[i];
  for (LL i = n; i < sz; i++) a[i] = 0;
  for (LL i = m; i < sz; i++) b[i] = 0;
  NTT(a, sz); NTT(b, sz);
  for (LL i = 0; i < sz; i++) a[i] = (a[i] * b[i]) % MOD
  ;
  NTT(a, sz, 1);
  vector<LL> c(a, a + sz);
  while (c.size() && c.back() == 0) c.pop_back();
  return c;
}
```

3.13 primality test

```
using u64 = uint64_t;
using u128 = __uint128_t;
// u64 binpower(u64 base, u64 e, u64 mod)
bool check_composite(u64 n, u64 a, u64 d, int s) {
  u64 x = binpower(a, d, n);
  if (x == 1 || x == n - 1) return false;
  for (int r = 1; r < s; r++) {
```

```
    x = (u128)x * x % n;
    if (x == n - 1) return false;
  }
  return true;
};
bool MillerRabin(u64 n, int iter = 5) { // returns true
if n is probably prime, else returns false.
  if (n < 4) return n == 2 || n == 3;
  int s = 0; u64 d = n - 1;
  while ((d & 1) == 0) d >>= 1, s++;
  for (int i = 0; i < iter; i++) {
    int a = 2 + rand() % (n - 3);
    if (check_composite(n, a, d, s)) return false;
  } return true;
}
```

4 Graph

4.1 BridgeTree

```
vector<PLL> g[N];
vector<int> ng[N];
int disc[N], low[N], mark[N], vis[N], timer = 1;
void find_bridge(int u, int p) {
  disc[u] = low[u] = timer++;
  bool fl = 1;
  for (auto [v, id] : g[u]) {
    if (v == p && fl) {
      fl = 0; continue;
    }
    if (disc[v]) {
      low[u] = min(low[u], disc[v]);
    } else {
      find_bridge(v, u);
      low[u] = min(low[u], low[v]);
      if (disc[u] < low[v]) mark[id] = 1;
    }
  }
}
void colorComponents(int u, int color) {
  if (vis[u]) return;
  vis[u] = color;
  for (auto [v, id] : g[u]) {
    if (mark[id]) continue;
    colorComponents(v, color);
  }
}
void solve() {
  int n, m;
  cin >> n >> m;
  vector<PLL> edges;
  for (int i = 0; i < m; i++) {
    int u, v; cin >> u >> v;
    edges.push_back({u, v});
    g[u].push_back({v, i});
    g[v].push_back({u, i});
  }
  find_bridge(1, 0);
  int color = 1;
  for (int i = 1; i <= n; i++) {
    if (!vis[i]) colorComponents(i, color++);
  }
  for (int i = 0; i < m; i++) {
    if (mark[i]) {
      ng[vis[edges[i].first]].push_back(vis[edges[i].
      second]);
      ng[vis[edges[i].second]].push_back(vis[edges[i].
      first]);
    }
  }
}
```

4.2 CD

```
class CD {
  vector<vector<int>> adj;
  vector<int> sub;
  vector<bool> blocked;
  int N;
 public:
  CD(vector<vector<int>> adj) : adj(adj) {
    N = tree.adj.size() - 1;
    blocked.assign(N + 1, 0);
    sub.assign(N + 1, 0);
    compute();
  }
  void compute(int u = 1, int p = 0) {
    sub[u] = 1;
    for (auto v : adj[u])
      if (v != p) {
        compute(v, u);
        sub[u] += sub[v];
      }
  }
  int centroid(int u, int p = 0) {
    int tot = sub[u];
    for (auto v : adj[u]) {
      if (v == p || blocked[v]) continue;
      if (2 * sub[v] > tot) {
        sub[u] = tot - sub[v];
        sub[v] = tot;
        return centroid(v, u);
      }
    }
    return u;
  }
  int count(int u, int p) { /* calculate ans */ }
  void update(int u, int p) { /* update */ }
  int decompose(int u = 1) {
    u = centroid(u);
    blocked[u] = 1;
    int ans = 0;
    /////  Do something here  //// count() update()
    for (auto v : adj[u])
      if (!blocked[v]) {
        ans += count(v, u);
        update(v, u);
      }
    /// reset updates here
    for (auto v : adj[u])
      if (!blocked[v])
        decompose(v);
    return ans;
  }
};
```

4.3 DSU with rollback

```
struct Rollback_DSU {
  int n;
  vector <int> par, sz;
  vector <pair <int, int>> op;
  Rollback_DSU(int n) : par(n), sz(n, 1) {
    iota(par.begin(), par.end(), 0);
    op.reserve(n);
  }
  int Anc(int node) {
    for(; node != par[node]; node = par[node]);
    return node;
  }
  bool Unite(int x, int y) {
```

```
    x = Anc(x), y = Anc (y);
    if (x == y) return false;
    if(sz[x] < sz[y]) swap(x, y);
    op.emplace_back(x, y);
    par[y] = x;
    sz[x] += sz[y];
    return true;
  }
  void Undo(size_t t) {
    for(; op.size() > t; op.pop_back()) {
      par[op.back().second] = op.back().second;
      sz[op.back().first] -= sz[op.back().second];
    }
  }
};
```

4.4 DSU, MST

```
class DSU {
  vector<int> parent, size;

 public:
  DSU(int n) : parent(n + 1), size(n + 1, 1) {
    iota(parent.begin(), parent.end(), 0);
  }
  int root(int u) {
    if (parent[u] == u) return u;
    return parent[u] = root(parent[u]);
  }
  bool same(int u, int v) { return root(u) == root(v); }
  void merge(int u, int v) {
    u = root(u), v = root(v);
    if (u == v) return;
    if (size[u] < size[v]) swap(u, v);
    parent[v] = u, size[u] += size[v];
  }
};
int kruskal(vector<tuple<int, int, int>> edges, int n) {
  sort(edges.begin(), edges.end());
  DSU mst(n);
  int cost = 0;
  for (auto& [w, u, v] : edges) {
    if (mst.same(u, v)) continue;
    mst.merge(u, v);
    cost += w;
  }
  return cost;
} /* PRIM'S SPANNING TREE (MST) */
DIJKSTRA code... start from a node,
    and push nodes which are not marked popped edges
    weight are taken
```

4.5 Dinic

```
struct Dinic {
  struct edge {
    int to, rid, isRev;
    LL cap, flow;
  };
  int n, s, t;
  vector<vector<edge>> g;
  vector<int> level, ptr;
  Dinic(int _n) : n(_n + 5) { g.resize(n); }
  void addEdge(int u, int v, LL cap) {
    g[u].push_back({v, (int)g[v].size(), 0, cap, 0LL});
    g[v].push_back({u, (int)g[u].size() - 1, 1, 0LL, 0LL
    });
  }
  bool bfs() {
    level.assign(n, -1);
    queue<int> q;
    q.push(s);
    level[s] = 0;
    while (!q.empty()) {
      int u = q.front();
      q.pop();
      for (auto& e : g[u]) {
        int v = e.to;
        if (level[v] == -1 and e.cap > e.flow)
          level[v] = level[u] + 1, q.push(v);
      }
    }
    return level[t] != -1;
  }
  LL dfs(int u, LL flow) {
    if (u == t) return flow;
    for (int& i = ptr[u]; i < (int)g[u].size(); i++) {
      edge& e = g[u][i];
      int v = e.to;
      if (level[v] != level[u] + 1 or e.cap <= e.flow)
      continue;
      LL cflow = dfs(v, min(flow, e.cap - e.flow));
      if (cflow) {
        e.flow += cflow, g[v][e.rid].flow -= cflow;
        return cflow;
      }
    }
    return 0;
  }
  LL maxFlow(int s, int t) {
    this->s = s, this->t = t;
    LL flow = 0;
    while (bfs()) {
      ptr.assign(n, 0);
      while (LL cflow = dfs(s, INF)) flow += cflow;
    }
    return flow;
  }
};
```

4.6 HLD

```
const int N = 2E5 + 69;
int n;
// seg tree here
vector <int> adj[N];
int cur_pos;
bool WEIGHT_IN_EDGE = false;
int Parent[N], Depth[N], Heavy[N], Head[N], Pos[N];
int Start[N], End[N]; vector <int> Euler; /// ONLY
NEEDED FOR SUBTREE QUERIES
void reset (bool WIE = false) {
  cur_pos = 0;
  WEIGHT_IN_EDGE = WIE;
  fill (Heavy, Heavy + n, -1);
  // build (vector <T> (n, I));
}
int dfs (int v = 0) {
  int size = 1, max_c_size = 0;
  for (int c : adj[v]) {
    if (c != Parent[v]) {
      Parent[c] = v;
      Depth[c] = Depth[v] + 1;
      int c_size = dfs(c);
      size += c_size;
      if (c_size > max_c_size) {
        max_c_size = c_size;
        Heavy[v] = c;
  } } }
  return size;
}
```

```
void decompose (int v = 0, int h = 0) {
  Head[v] = h; Pos[v] = cur_pos++;
  Euler.push_back(v);
  Start[v] = (int)Euler.size() - 1;
  if (Heavy[v] != -1) decompose(Heavy[v], h);
  for (int c : adj[v]) if (c != Parent[v] && c != Heavy[
  v]) decompose(c, c);
  End[v] = Euler.size() - 1;
}
void update_path (int a, int b, LL val) {
  while (Head[a] != Head[b]) {
    if (Depth[Head[a]] < Depth[Head[b]]) swap(a, b);
    update(Pos[Head[a]], Pos[a], val);
    a = Parent[Head[a]];
  }
  if (Depth[a] > Depth[b]) swap(a, b);
  update(Pos[a] + WEIGHT_IN_EDGE, Pos[b], val);
}
// LL query_path (int a, int b) {
//   LL res = 0;
//   while (Head[a] != Head[b]) {
//     if (Depth[Head[a]] < Depth[Head[b]]) swap(a, b);
//     res = merge (res, query(Pos[Head[a]], Pos[a]));
//     a = Parent[Head[a]];
//   }
//   if (Depth[a] > Depth[b]) swap(a, b);
//   res = merge (res, query(Pos[a] + WEIGHT_IN_EDGE, Pos
[b]));
//   return res;
// }
vector <T> query_up(int u, int anc) {
  vector <T> res;
  while (Head[u] != Head[anc]) {
    res.emplace_back (query(Pos[Head[u]], Pos[u]));
    u = Parent[Head[u]];
  }
  res.emplace_back (query(Pos[anc] + 1, Pos[u]));
  return res;
}
LL query_path(int u, int v) {
  int w = LCA:: lca(u, v);
  auto left = query_up(u, w);
  auto right = query_up(v, w);
  node res = I;
  for (auto e : left) res = merge (e, res);
  if (not WEIGHT_IN_EDGE) res = merge(query(Pos[w], Pos[
  w]), res);
  swap (res.AP, res.PA);
  auto res2 = I;
  for (auto e : right) res2 = merge (e, res2);
  res = merge (res, res2);
  return max ({res.PP, res.PA, res.AP, res.AA});
}
void update_subtree (int v, LL val) { update(Start[v],
End[v], val); }
LL query_subtree (int v) { return query(Start[v], End[v
]); }
void HLD_initialize (vector<LL> &weight) {
  reset (true); Parent[0] = -1;
  dfs(); decompose();
  for (int i = 0; i < n; i++) update(Pos[i], Pos[i],
  weight[i]);
}
```

4.7 LCA O(n)

```
const int K = lg(N) + 1;
vector<int> adj[N];
int anc[N][K], lvl[N];
namespace lca {
  void init(int u = 1, int p = 0, int d = 0) {
    lvl[u] = d;
    anc[u][0] = p;
    for (int i = 1; i < K; i++)
      anc[u][i] = anc[anc[u][i - 1]][i - 1];
    for (auto v : adj[u])
      if (v != p)
        init(v, u, d + 1);
  }
  int getAnc(int u, int k) {
    for (int i = 0; u and i < K; i++)
      if ((k >> i) & 1)
        u = anc[u][i];
    return u;
  }
  int lca(int u, int v) {
    if (lvl[u] < lvl[v]) swap(u, v);
    u = getAnc(u, lvl[u] - lvl[v]);
    if (u == v) return u;
    for (int i = K - 1; ~i; i--)
      if (anc[u][i] != anc[v][i])
        u = anc[u][i], v = anc[v][i];
    return anc[u][0];
  }
  int dist(int u, int v) {
    return lvl[u] + lvl[v] - 2 * lvl[lca(u, v)];
  }
};
```

4.8 TwoSat

```
struct TwoSatSolver{
    int n_var;
    vector<vector<int>> adj, radj;
    vector<int> order, vis, comp, assignment;
    void dfs(int u){
        if(vis[u]) return;
        vis[u] = 1;
        for(auto v: adj[u])
            dfs(v);
        order.push_back(u);
    }
    void asgn(int u, int id){
        if(comp[u]) return;
        comp[u] = id;
        for(auto v: radj[u])
            asgn(v, id);
    }
    TwoSatSolver(int n) : n_var(n + 1), adj(2 * n_var),
    radj(2 * n_var), vis(2 * n_var), comp(2 * n_var),
    assignment(n_var) {
        order.reserve(2 * n_var);
    }
    void add(int a, int nega, int b, int negb){
        a = 2 * a ^ nega, b = 2 * b ^ negb;
        adj[a ^ 1].push_back(b);
        radj[b].push_back(a ^ 1);
        adj[b ^ 1].push_back(a);
        radj[a].push_back(b ^ 1);
    }
    bool isSatisfiable(){
        order.clear();
        for(int i = 0; i < 2 * n_var; i++){
            if(!vis[i]) dfs(i);
        }
        reverse(all(order));
        for(int i = 0, j = 1; i < 2 * n_var; i++){
            if(!comp[order[i]]) asgn(order[i], j++);
        }
```

```
        bool fl = 1;
        for(int i = 0; i < n_var; i++){
            int l = 2 * i, r = 2 * i + 1;
            if(comp[l] == comp[r]) fl = 0;
            if(comp[l] < comp[r]) assignment[i] = 1;
        }
        return fl;
    }
};
```

4.9 block cut tree

```
const int N = 4e5 + 1;
bitset<N> art;
vector<int> g[N], tree[N], st, comp[N];
int n, ptr, cur, in[N], low[N], id[N];
void memclear() {
  for (int i = 1; i <= n; i++) {
    g[i].clear(); in[i] = low[i] = art[i] = 0;
  }
  for (int i = 1; i <= ptr; i++) {
    tree[i].clear(); comp[i].clear();
  }
  st.clear();
  ptr = cur = 0;
}
void dfs(int u, int from = -1) {
  in[u] = low[u] = ++ptr;
  st.emplace_back(u);
  for (int v : g[u])
    if (v ^ from) {
      if (!in[v]) {
        dfs(v, u); low[u] = min(low[u], low[v]);
        if (low[v] >= in[u]) {
          art[u] = ~from or in[v] > in[u] + 1;
          comp[++cur].emplace_back(u);
          while (comp[cur].back() ^ v) {
            comp[cur].emplace_back(st.back()); st.
            pop_back();
          } }
      } else low[u] = min(low[u], in[v]);
    }
}
void build_tree() {
  ptr = 0;
  for (int i = 1; i <= n; ++i)
    if (art[i]) id[i] = ++ptr;
  for (int i = 1; i <= cur; ++i) {
    int x = ++ptr;
    for (int u : comp[i]) {
      if (art[u]) {
        tree[x].emplace_back(id[u]); tree[id[u]].
        emplace_back(x);
      } else id[u] = x;
    } }
}
```

4.10 maximum bipartite matching hopkroft

```
// do everything 1-based
class BipartiteMatcher {
 private:
  int n, m;              // Number of vertices in left
  and right sets
  vector<vector<int>> adj; // Adjacency list for the
  bipartite graph
  vector<int> dist;      // Distance array for BFS
  vector<int> matchL, matchR;
  // matchL[u] is the right vertex matched with u;
  // matchR[v] is the left vertex matched with v
 public:
  BipartiteMatcher(int n, int m) : n(n), m(m) {
    adj.resize(n + 1);
    matchL.resize(n + 1, 0);
    matchR.resize(m + 1, 0);
    dist.resize(n + 1);
  }
  void addEdge(int u, int v) { adj[u].push_back(v); }
  bool bfs() {
    queue<int> q;
    for (int u = 1; u <= n; u++) {
      if (matchL[u] == 0) {
        dist[u] = 0;
        q.push(u);
      } else {
        dist[u] = INT_MAX;
      }
    }
    dist[0] = INT_MAX;
    while (!q.empty()) {
      int u = q.front(); q.pop();
      if (dist[u] < dist[0]) {
        for (int v : adj[u]) {
          if (dist[matchR[v]] == INT_MAX) {
            dist[matchR[v]] = dist[u] + 1;
            q.push(matchR[v]);
          }
        }
      }
    }
    return dist[0] != INT_MAX;
  }
  bool dfs(int u) {
    if (u != 0) {
      for (int v : adj[u]) {
        if (dist[matchR[v]] == dist[u] + 1 and dfs(matchR
        [v])) {
          matchL[u] = v; matchR[v] = u;
          return true;
        }
      }
      dist[u] = INT_MAX;
      return false;
    }
    return true;
  }
  int hopcroftKarp() {
    int matching = 0;
    while (bfs())
      for (int u = 1; u <= n; u++)
        if (matchL[u] == 0 && dfs(u)) matching++;
    return matching;
  }
};
```

5 String

5.1 Aho

```
const int N = 2000, L = 105;
struct AhoCorasick {
  int N, P;
  const int A = 256;
  vector<vector<int>> next;
  vector<int> link, out_link, end_in_pattern;
  vector<vector<int>> out;
  AhoCorasick() : N(0), P(0) { node(); }
  int node() {
    next.emplace_back(A, 0);
    link.emplace_back(0);
    out_link.emplace_back(0);
```

```
    out.emplace_back(0);
    end_in_pattern.emplace_back(0);
    return N++;
  }
  inline int get(char c) { return c; }
  int addPattern(const string T) {
    int u = 0;
    for (auto c : T) {
      if (!next[u][get(c)]) next[u][get(c)] = node();
      u = next[u][get(c)];
    }
    out[u].push_back(P);
    end_in_pattern[u] = 1;
    return P++;
  }
  void pushLinks() {
    queue<int> q;
    for (q.push(0); !q.empty();) {
      int u = q.front();
      q.pop();
      for (int c = 0; c < A; ++c) {
        int v = next[u][c];
        if (!v) next[u][c] = next[link[u]][c];
        else {
          link[v] = u ? next[link[u]][c] : 0;
          out_link[v] = out[link[v]].empty() ? out_link[
          link[v]] : link[v];
          q.push(v);
        }
        end_in_pattern[v] |= end_in_pattern[out_link[v]];
  } } }
  int advance(int u, char c) {
    while (u && !next[u][get(c)]) u = link[u];
    u = next[u][get(c)]; return u;
} };
```

5.2 KMP

```
template <typename T>
vector<int> getLPSarray(T& b) {
  int m = b.size();
  vector<int> phi(m);
  for (int i = 1, j = 0; i < m; i++) {
    while (j && b[i] != b[j]) j = phi[j - 1];
    if (b[i] == b[j]) j++;
    phi[i] = j;
  } return phi;
}
template <typename T>
int KMP_match(T& a, T& b) {
  int n = a.size(), m = b.size();
  vector<int> lps = getLPSarray(b);
  int i = 0, j = 0, cnt = 0;
  while (n - i >= m - j) {
    if (a[i] == b[j]) i++, j++;
    if (j == m) cnt++, j = lps[j - 1];
    else if (i < n && a[i] != b[j]) {
      if (j) j = lps[j - 1];
      else i++;
  } }
  return cnt;
}
```

5.3 Manacher

```
void Manacher() {
  vector<int> d1(n);
  // d[i] = number of palindromes taking s[i] as center
  for (int i = 0, l = 0, r = -1; i < n; i++) {
    int k = (i > r) ? 1 : min(d1[l + r - i], r - i + 1);
    while (0 <= i - k && i + k < n && s[i - k] == s[i +
    k]) k++;
    d1[i] = k--;
    if (i + k > r) l = i - k, r = i + k;
  }
  vector<int> d2(n);
  // d[i] = number of palindromes taking s[i-1] and s[i]
   as center
  for (int i = 0, l = 0, r = -1; i < n; i++) {
    int k = (i > r) ? 0 : min(d2[l + r - i + 1], r - i +
     1);
    while (0 <= i - k - 1 && i + k < n && s[i - k - 1]
    == s[i + k]) k++;
    d2[i] = k--;
    if (i + k > r) l = i - k - 1, r = i + k;
} }
```

5.4 String Match FFT

```
/*find occurrences of t in s where '?'s are
automatically matched with any
 * character res[i + m - 1] = sum_j=0 to m - 1_{s[i + j]
  * t[j] * (s[i + j] -
 * t[j]) */
vector<int> string_matching(string& s, string& t) {
  int n = s.size(), m = t.size();
  vector<int> s1(n), s2(n), s3(n);
  for (int i = 0; i < n; i++)
    s1[i] = s[i] == '?' ? 0 : s[i] - 'a' + 1; /*assign
    any non zero number for non '?'s */
  for (int i = 0; i < n; i++) s2[i] = s1[i] * s1[i];
  for (int i = 0; i < n; i++) s3[i] = s1[i] * s2[i];
  vector<int> t1(m), t2(m), t3(m);
  for (int i = 0; i < m; i++) t1[i] = t[i] == '?' ? 0 :
  t[i] - 'a' + 1;
  for (int i = 0; i < m; i++) t2[i] = t1[i] * t1[i];
  for (int i = 0; i < m; i++) t3[i] = t1[i] * t2[i];
  reverse(t1.begin(), t1.end());
  reverse(t2.begin(), t2.end());
  reverse(t3.begin(), t3.end());
  vector<int> s1t3 = multiply(s1, t3);
  vector<int> s2t2 = multiply(s2, t2);
  vector<int> s3t1 = multiply(s3, t1);
  vector<int> res(n);
  for (int i = 0; i < n; i++) res[i] = s1t3[i] - s2t2[i]
   * 2 + s3t1[i];
  vector<int> oc;
  for (int i = m - 1; i < n; i++)
    if (res[i] == 0) oc.push_back(i - m + 1);
  return oc;
}
```

5.5 Z

```
vector<int> z(string const& s) {
  int n = size(s);
  vector<int> z(n);
  int x = 0, y = 0;
  for (int i = 1; i < n; i++) {
    z[i] = max(0, min(z[i - x], y - i + 1));
    while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
      x = i, y = i + z[i], z[i]++;
  } } return z;
}
```

5.6 double hashing

```
// Some Primes : 1000000007, 1000000009, 1000000861,
1000099999(< 2 ^ 30)1088888881, 1111211111, 1500000001,
1481481481(< 2 ^ 31)
namespace Hashing {
#define ff first
#define ss second
```

```
const PLL M = {1e9 + 7, 1e9 + 9};
const LL base = 1259; const int N = 1e6 + 7;
PLL operator+(const PLL& a, LL x) { return {a.ff + x, a.
ss + x}; }
PLL operator-(const PLL& a, LL x) { return {a.ff - x, a.
ss - x}; }
PLL operator*(const PLL& a, LL x) { return {a.ff * x, a.
ss * x}; }
PLL operator+(const PLL& a, PLL x) { return {a.ff + x.ff
, a.ss + x.ss}; }
PLL operator-(const PLL& a, PLL x) { return {a.ff - x.ff
, a.ss - x.ss}; }
PLL operator*(const PLL& a, PLL x) { return {a.ff * x.ff
, a.ss * x.ss}; }
PLL operator%(const PLL& a, PLL m) { return {a.ff % m.ff
, a.ss % m.ss}; }
ostream& operator<<(ostream& os, PLL hash) {
  return os << "(" << hash.ff << ", " << hash.ss << ")";
}
PLL pb[N];
void hashPre() { pb[0] = {1, 1}; for (int i = 1; i < N;
i++) pb[i] = (pb[i - 1] * base) % M; }
vector<PLL> hashList(string s) {
  int n = s.size(); vector<PLL> ans(n + 1); ans[0] = {0,
   0};
  for (int i = 1; i <= n; i++) ans[i] = (ans[i - 1] *
  base + s[i - 1]) % M;
  return ans;
}
PLL substringHash(const vector<PLL>& hashlist, int l,
int r) {
  return (hashlist[r] + (M - hashlist[l - 1]) * pb[r - l
   + 1]) % M;
}
PLL Hash(string s) { PLL ans = {0, 0}; for (int i = 0; i
 < s.size(); i++) ans = (ans * base + s[i]) % M; return
ans; }
PLL append(PLL cur, char c) { return (cur * base + c) %
M; }
PLL prepend(PLL cur, int k, char c) { return (pb[k] * c
+ cur) % M; } ///replaces the i-th (0-indexed) character
 from right from a to b;
PLL replace(PLL cur, int i, char a, char b) { return (
cur + pb[i] * (M + b - a)) % M; } ///Erases c from front
 of the string with size len
PLL pop_front(PLL hash, int len, char c) { return (hash
+ pb[len - 1] * (M - c)) % M; } ///concatenates two
strings where length of the right is k
PLL concat(PLL left, PLL right, int k) { return (left *
pb[k] + right) % M; }
PLL inverse(PLL a) {
  if (M.ss == 1) return bigmod(a, M.ff - 2);
  return bigmod(a, (M.ff - 1) * (M.ss - 1) - 1);
} ///Erases c from the back of the string
PLL invb = inverse({base, base});
PLL pop_back(PLL hash, char c) { return ((hash - c + M)
* invb) % M; } ///Calculates hash of string with size
len repeated cnt times
///This is O(log n). For O(1), pre-calculate inverses
PLL repeat(PLL hash, int len, LL cnt) {
  PLL mul = ((pb[len * cnt] - 1 + M) * inverse(pb[len] -
   1 + M)) % M;
  PLL ans = (hash * mul);
  if (pb[len].ff == 1) ans.ff = hash.ff * cnt;
  if (pb[len].ss == 1) ans.ss = hash.ss * cnt;
  return ans % M;
} }
```

5.7 suffix array

```
const int N = 3e5 + 9;
const int LG = 18;
void induced_sort(const vector<int> &vec, int val_range,
 vector<int> &SA, const vector<bool> &sl, const vector<
int> &lms_idx) {
  vector<int> l(val_range, 0), r(val_range, 0);
  for (int c : vec) {
    if (c + 1 < val_range) ++l[c + 1];
    ++r[c];
  }
  partial_sum(l.begin(), l.end(), l.begin());
  partial_sum(r.begin(), r.end(), r.begin());
  fill(SA.begin(), SA.end(), -1);
  for (int i = lms_idx.size() - 1; i >= 0; --i)
    SA[--r[vec[lms_idx[i]]]] = lms_idx[i];
  for (int i : SA) if (i >= 1 && sl[i - 1])
    SA[l[vec[i - 1]]++] = i - 1;
  fill(r.begin(), r.end(), 0);
  for (int c : vec) ++r[c];
  partial_sum(r.begin(), r.end(), r.begin());
  for (int k = SA.size() - 1, i = SA[k]; k >= 1; --k, i
  = SA[k])
    if (i >= 1 && !sl[i - 1])
      SA[--r[vec[i - 1]]] = i - 1;
}
vector<int> SA_IS(const vector<int> &vec, int val_range)
 {
  const int n = vec.size();
  vector<int> SA(n), lms_idx;
  vector<bool> sl(n);
  sl[n - 1] = false;
  for (int i = n - 2; i >= 0; --i) {
    sl[i] = (vec[i] > vec[i + 1] || (vec[i] == vec[i +
    1] && sl[i + 1]));
    if (sl[i] && !sl[i + 1]) lms_idx.push_back(i + 1);
  }
  reverse(lms_idx.begin(), lms_idx.end());
  induced_sort(vec, val_range, SA, sl, lms_idx);
  vector<int> new_lms_idx(lms_idx.size()), lms_vec(
  lms_idx.size());
  for (int i = 0, k = 0; i < n; ++i)
    if (!sl[SA[i]] && SA[i] >= 1 && sl[SA[i] - 1])
      new_lms_idx[k++] = SA[i];
  int cur = 0;
  SA[n - 1] = cur;
  for (size_t k = 1; k < new_lms_idx.size(); ++k) {
    int i = new_lms_idx[k - 1], j = new_lms_idx[k];
    if (vec[i] != vec[j]) {
      SA[j] = ++cur;
      continue;
    }
    bool flag = false;
    for (int a = i + 1, b = j + 1;; ++a, ++b) {
      if (vec[a] != vec[b]) {
        flag = true; break;
      }
      if ((!sl[a] && sl[a - 1]) || (!sl[b] && sl[b - 1]))
       {
        flag = !((!sl[a] && sl[a - 1]) && (!sl[b] && sl[b
         - 1])); break;
      }
    }
    SA[j] = (flag ? ++cur : cur);
  }
  for (size_t i = 0; i < lms_idx.size(); ++i)
    lms_vec[i] = SA[lms_idx[i]];
  if (cur + 1 < (int)lms_idx.size()) {
    auto lms_SA = SA_IS(lms_vec, cur + 1);
```

```
    for (size_t i = 0; i < lms_idx.size(); ++i) {
      new_lms_idx[i] = lms_idx[lms_SA[i]];
    }
  }
  induced_sort(vec, val_range, SA, sl, new_lms_idx);
  return SA;
}
vector<int> suffix_array(const string &s, const int LIM
= 128) {
  vector<int> vec(s.size() + 1);
  copy(begin(s), end(s), begin(vec));
  vec.back() = '$';
  auto ret = SA_IS(vec, LIM);
  ret.erase(ret.begin());
  return ret;
}
struct SuffixArray {
  int n; string s;
  vector<int> sa, rank, lcp;
  vector<vector<int>> t;
  vector<int> lg;
  SuffixArray() {}
  SuffixArray(string _s) {
    n = _s.size();
    s = _s;
    sa = suffix_array(s);
    rank.resize(n);
    for (int i = 0; i < n; i++) rank[sa[i]] = i;
    costruct_lcp();
    prec();
    build();
  }
  void costruct_lcp() {
    int k = 0;
    lcp.resize(n - 1, 0);
    for (int i = 0; i < n; i++) {
      if (rank[i] == n - 1) {
        k = 0; continue;
      }
      int j = sa[rank[i] + 1];
      while (i + k < n && j + k < n && s[i + k] == s[j +
      k]) k++;
      lcp[rank[i]] = k;
      if (k) k--;
  } }
  void prec() {
    lg.resize(n, 0);
    for (int i = 2; i < n; i++) lg[i] = lg[i / 2] + 1;
  }
  void build() {
    int sz = n - 1;
    t.resize(sz);
    for (int i = 0; i < sz; i++) {
      t[i].resize(LG); t[i][0] = lcp[i];
    }
    for (int k = 1; k < LG; ++k) {
      for (int i = 0; i + (1 << k) - 1 < sz; ++i) {
        t[i][k] = min(t[i][k - 1], t[i + (1 << (k - 1))][
        k - 1]);
  } } }
  int query(int l, int r) { // minimum of lcp[l], ...,
  lcp[r]
    int k = lg[r - l + 1];
    return min(t[l][k], t[r - (1 << k) + 1][k]);
  }
  int get_lcp(int i, int j) { // lcp of suffix starting
  from i and j
    if (i == j) return n - i;
    int l = rank[i], r = rank[j];
    if (l > r) swap(l, r);
    return query(l, r - 1);
  }
  int lower_bound(string &t) {
    int l = 0, r = n - 1, k = t.size(), ans = n;
    while (l <= r) {
      int mid = l + r >> 1;
      if (s.substr(sa[mid], min(n - sa[mid], k)) >= t)
      ans = mid, r = mid - 1;
      else l = mid + 1;
    } return ans;
  }
  int upper_bound(string &t) {
    int l = 0, r = n - 1, k = t.size(), ans = n;
    while (l <= r) {
      int mid = l + r >> 1;
      if (s.substr(sa[mid], min(n - sa[mid], k)) > t) ans
       = mid, r = mid - 1;
      else l = mid + 1;
    } return ans;
  }
  // occurrences of s[p, ..., p + len - 1]
  pair<int, int> find_occurrence(int p, int len) {
    p = rank[p];
    pair<int, int> ans = {p, p};
    int l = 0, r = p - 1;
    while (l <= r) {
      int mid = l + r >> 1;
      if (query(mid, p - 1) >= len) ans.first = mid, r =
      mid - 1;
      else l = mid + 1;
    }
    l = p + 1, r = n - 1;
    while (l <= r) {
      int mid = l + r >> 1;
      if (query(p, mid - 1) >= len) ans.second = mid, l =
       mid + 1;
      else r = mid - 1;
    } return ans;
} };
```

5.8 $\text{suffix}_{array_n log n}$

```
vector<int> getSuffixArray(string s){
    s.push_back(*min_element(all(s)) - 1);
    vector<int> values(all(s));
    sort(all(s)); s.erase(unique(all(s)), s.end());
    for(auto &u: values) u = lower_bound(all(s), u) - s.
    begin();
    int n = values.size();
    vector<int> order(n); iota(all(order), 0);
    auto c_sort = [&](int k, int bckt){
        vector<int> norder(n), freq(bckt + 1);
        for(int i = 0; i < n; i++) norder[i] = order[i]
        >= k ? order[i] - k : order[i] + n - k;
        for(auto &u: norder) freq[values[u]]++;
        partial_sum(all(freq), freq.begin());
        for(int i = n - 1; i >= 0; i--) order[--freq[
        values[norder[i]]]] = norder[i];
    };
    c_sort(0, n - 1);
    for(int k = 1; k < n; k *= 2){
        c_sort(k, values[order[n - 1]]);
        vector<int> nvalues(n);
        for(int u = 1; u < n; u++){
            int i = order[u - 1], j = order[u];
            nvalues[j] = nvalues[i];
            if(values[i] != values[j] or values[(i + k) %
             n] != values[(j + k) % n]) nvalues[j]++;
        }
```

```
        values = nvalues;
    }
    order.erase(order.begin());
    return order;
}
vector<int> getLCParray(string &s, vector<int> &sa){
    int n = sa.size();
    vector<int> lcp(n), rank(n);
    for(int i = 0; i < n; i++) rank[sa[i]] = i;
    for(int i = 0, k = 0; i < n; lcp[rank[i]] = k, i++,
    k ? --k : 0){
        if(rank[i] == n - 1) { k = 0; continue; }
        int j = sa[rank[i] + 1];
        while(i + k < n and j + k < n and s[i + k] == s[j
         + k]) k++;
    }
    return lcp;
}
```

6 DP

6.1 CHT

```
struct Line {
  mutable LL m, c, p;
  bool operator<(const Line& o) const { return m < o.m;
  }
  bool operator<(LL x) const { return p < x; }
};
/* this calculates maximum value of m * x + c over all
lines */
/* to get minimum value use m = -m , c = -c , query(x) =
 -query(x) */
struct LineContainer : multiset<Line, less<>> {
  /* (for doubles, use inf = 1/.0, div(a,b) = a/b) */
  static const LL inf = LLONG_MAX;
  LL div(LL a, LL b) { /* floored division */
    return a / b - ((a ^ b) < 0 && a % b);
  }
  bool isect(iterator x, iterator y) {
    if (y == end()) return x->p = inf, 0;
    if (x->m == y->m)
      x->p = x->c > y->c ? inf : -inf;
    else
      x->p = div(y->c - x->c, x->m - y->m);
    return x->p >= y->p;
  }
  void add(LL m, LL c) {
    auto z = insert({m, c, 0}), y = z++, x = y;
    while (isect(y, z)) z = erase(z);
    if (x != begin() && isect(--x, y)) isect(x, y =
    erase(y));
    while ((y = x) != begin() && (--x)->p >= y->p) isect
    (x, erase(y));
  }
  LL query(LL x) {
    assert(!empty());
    auto l = *lower_bound(x);
    return l.m * x + l.c;
  }
};
```

6.2 CatalanDp

```
// comb formula: ((2n)Cn)-((2n)C(n-1)) = (1/(n+1))*((2n)
Cn)
void genCatalan(int n) {
  catalan[0] = catalan[1] = 1;
  for (int i = 2; i <= n; i++) {
    catalan[i] = 0;
    for (int j = 0; j < i; j++) {
      catalan[i] += (catalan[j] * catalan[i - j - 1]) %
      mod;
      if (catalan[i] >= mod) {
        catalan[i] -= mod;
      } } }
}
```

6.3 DearrangementDP

```
void gen_drng(int n) {
  drng[2] = 1;
  for (int i = 3; i <= n; i++)
    drng[i] = ((i - 1ll) * ((drng[i - 2] + drng[i - 1])
    % mod)) % mod;
}
```

6.4 Li Chao Tree

```
struct Line {
  double m, c;
  double eval(double x) { return m * x + c; }
};
struct node {
  Line line;
  node* left = nullptr;
  node* right = nullptr;
  node(Line line) : line(line) {}
  void add_segment(Line nw, LL l, LL r, LL L, LL R) {
    if (l > r || r < L || l > R) return;
    LL m = (l + 1 == r ? l : (l + r) / 2);
    if (l >= L and r <= R) {
      bool lef = nw.eval(l) < line.eval(l);
      bool mid = nw.eval(m) < line.eval(m);
      if (mid) swap(line, nw);
      if (l == r) return;
      if (lef != mid) {
        if (left == nullptr)
          left = new node(nw);
        else
          left->add_segment(nw, l, m, L, R);
      } else {
        if (right == nullptr)
          right = new node(nw);
        else
          right->add_segment(nw, m + 1, r, L, R);
      }
      return;
    }
    if (max(l, L) <= min(m, R)) {
      if (left == nullptr) left = new node({0, inf});
      left->add_segment(nw, l, m, L, R);
    }
    if (max(m + 1, L) <= min(r, R)) {
      if (right == nullptr) right = new node({0, inf});
      right->add_segment(nw, m + 1, r, L, R);
    }
  }
  double query_segment(LL x, LL l, LL r, LL L, LL R) {
    if (l > r || r < L || l > R) return inf;

    double ans = line.eval(x);
    if (l == r) return ans;

    LL m = (l + r) >> 1;

    if (x <= m && left) ans = min(ans, left->
    query_segment(x, l, m, L, R));
    if (x > m && right) ans = min(ans, right->
    query_segment(x, m + 1, r, L, R));

    return ans;
```

```
  }
};
struct LiChaoTree {
  LL L, R;
  node* root;
  LiChaoTree(LL L, LL R) : L(L), R(R) { root = new node
  ({0, inf}); }
  void add_segment(Line line, LL l, LL r) {
    root->add_segment(line, L, R, l, r);
  }
  double query(LL x) { return root->query_segment(x, L,
  R, L, R); }
};

void solve(int tc) {
  int n;
  cin >> n;
  LiChaoTree tree(-INF, INF);
  for (int i = 0; i < n; i++) {
    LL x, v;
    cin >> x >> v;
    double m = -1.0 / v, c = 1.0 * x / v;
    tree.add_segment({m, c}, -INF, x);
    tree.add_segment({-m, -c}, x, INF);
  }
  int q;
  cin >> q;
  while (q--) {
    char t;
    cin >> t;
    if (t == '+') {
      LL x, v;
      cin >> x >> v;
      double m = -1.0 / v, c = 1.0 * x / v;
      tree.add_segment({m, c}, -INF, x);
      tree.add_segment({-m, -c}, x, INF);
    } else {
      LL x;
      cin >> x;
      cout << fixed << setprecision(9) << tree.query(x)
      << '\n';
    }
  }
}
```

6.5 SOS

```
for (int mask = 0; mask < (1 << N); ++mask) {
  dp[mask][-1] = A[mask];
  for (int i = 0; i < N; ++i) {
    if (mask & (1 << i))
      dp[mask][i] = dp[mask][i - 1] + dp[mask ^ (1 << i)
      ][i - 1];
    else
      dp[mask][i] = dp[mask][i - 1];
  }
  F[mask] = dp[mask][N - 1];
}
// good for inc-exc
for (int i = 0; i < (1 << N); ++i) F[i] = A[i];
for (int i = 0; i < N; ++i)
  for (int mask = 0; mask < (1 << N); ++mask)
    if (mask & (1 << i)) F[mask] += F[mask ^ (1 << i)];
```

6.6 grundy

```
/* single pile game-> greedy or game dp multiple pile
game and
 * disjunctive(before playing, choose 1 pile) -> NIM
 game else-> Grundy(converts
 * n any game piles to n NIM piles) grundy(x)->the
 smallest nonreachable grundy
 * value there are n pile of games and k type of moves.
 if XOR(grundy(games)) ==
 * 0: losing state else winning state */
vector<int> moves, dp;
int mex(vector<int>& a) {
  set<int> b(a.begin(), a.end());
  for (int i = 0;; ++i)
    if (!b.count(i)) return i;
}
int grundy(int x) {
  if (dp[x] != -1) return dp[x];
  vector<int> reachable;
  for (auto m : moves) {
    if (x - m < 0) continue;
    int val = grundy(x - m);
    reachable.push_back(val);
  }
  return dp[x] = mex(reachable);
}
```

6.7 knuth

```
// opt(i, j) = maximum value of k that minimizes dp(i, j
)
// C(i, j) = cost of single merging of (i, j)
// dp(i, j) = min cost of total merging of (i, j)
// necessary condition opt(i, j - 1) <= opt(i, j) <= opt
(i + 1, j)
// above condition is true if (not the only condition) C
(b, c) <= C(a, d) and C(a, c) + C(b, d) <= C(a, d) + C(b
, c) where a <= b <= c <= d
int solve() {
    int dp[N][N], opt[N][N];
    auto C = [&](int i, int j) {};
    for (int i = 0; i < N; i++) opt[i][i] = i;
    for (int i = N-2; i >= 0; i--) {
        for (int j = i+1; j < N; j++) {
            int mn = INT_MAX;
            int cost = C(i, j);
            for (int k = opt[i][j-1]; k <= min(j-1, opt[i
            +1][j]); k++) {
                if (mn >= dp[i][k] + dp[k+1][j] + cost) {
                    opt[i][j] = k;
                    mn = dp[i][k] + dp[k+1][j] + cost;
                }
            }
            dp[i][j] = mn;
        }
    }
    return dp[0][N-1];
}
```

7 Geometry

7.1 2D everything

```
using LL = long long;
using ULL = unsigned long long;
const double PI = acos(-1), EPS = 1e-10;
template <typename DT>
DT sq(DT x) {
  return x * x;
}
template <typename DT>
int dcmp(DT x) {
  return fabs(x) < EPS ? 0 : (x < 0 ? -1 : 1);
}
int sign(double x) { return (x > eps) - (x < -eps); }
```

```
template <typename DT>
class point {
 public:
  DT x, y;
  point() = default;
  point(DT x, DT y) : x(x), y(y) {};
  template <typename X>
  point(point<X> p) : x(p.x), y(p.y){};
  point operator+(const point& rhs) const {
    return point(x + rhs.x, y + rhs.y);
  }
  point operator-(const point& rhs) const {
    return point(x - rhs.x, y - rhs.y);
  }
  point operator*(const point& rhs) const {
    return point(x * rhs.x - y * rhs.y, x * rhs.y + y *
    rhs.x);
  }
  point operator/(const point& rhs) const {
    return *this * point(rhs.x, -rhs.y) / ~(rhs);
  }
  point operator*(DT M) const { return point(M * x, M *
  y); }
  point operator/(DT M) const { return point(x / M, y /
  M); }
  bool operator<(point rhs) const {
    return x < rhs.x or (x == rhs.x and y < rhs.y);
  }
  bool operator==(const point& rhs) const { return x ==
  rhs.x and y == rhs.y; }
  bool operator<=(const point& rhs) const {
    return *this < rhs or *this == rhs;
  }
  bool operator!=(const point& rhs) const { return x !=
  rhs.x or y != rhs.y; }
  DT operator&(const point& rhs) const {
    return x * rhs.y - y * rhs.x;
  } /* cross product */
  DT operator^(const point& rhs) const {
    return x * rhs.x + y * rhs.y;
  }                                          /* dot
  product */
  DT operator~() const { return sq(x) + sq(y); } /*
  square of norm */
  point operator-() const { return *this * -1; }
  friend istream& operator>>(istream& is, point& p) {
  return is >> p.x >> p.y; }
  friend ostream& operator<<(ostream& os, const point& p
  ) {
    return os << p.x << " " << p.y;
  }
  friend DT DisSq(const point& a, const point& b) {
    return sq(a.x - b.x) + sq(a.y - b.y);
  }
  friend DT TriArea(const point& a, const point& b,
  const point& c) {
    return (b - a) & (c - a);
  }
  friend DT UTriArea(const point& a, const point& b,
  const point& c) {
    return abs(TriArea(a, b, c));
  }
  friend bool Collinear(const point& a, const point& b,
  const point& c) {
    return UTriArea(a, b, c) < EPS;
  }
  friend DT orientation (const point &a, const point &b,
   const point &c) { return (b - a) & (c - a); }
  friend double Angle(const point& u) { return atan2(u.y
  , u.x); }
  friend double Angle(const point& a, const point& b) {
    double ans = Angle(b) - Angle(a);
    return ans <= -PI ? ans + 2 * PI : (ans > PI ? ans -
     2 * PI : ans);
  }
  friend double Angle(const point& A, const point& B,
  const point& C) {
    point BA = A - B;
    point BC = C - B;
    double dot = BA ^ BC;
    double magBA = sqrt(~BA);
    double magBC = sqrt(~BC);
    double cosTheta = dot / (magBA * magBC);
    cosTheta = std::max(-1.0, std::min(1.0, cosTheta));
    return acos(cosTheta);
  }
  point Perp(const point& a) { return point(-a.y, a.x);
  }
  point Conj(const point& a) { return point(a.x, -a.y);
  }
};
```

```
/* given a pivot point and an initial direction, sorts
by Angle with the given direction */
template <typename DT>
using polygon = vector<point<DT>>;
template <typename DT>
class polarComp {
  point<DT> O, dir;
  bool half(point<DT> p) {
    return dcmp(dir & p) < 0 || (dcmp(dir & p) == 0 &&
    dcmp(dir ^ p) > 0);
  }
 public:
  polarComp(point<DT> O = point(0, 0), point<DT> dir =
  point(1, 0))
      : O(O), dir(dir) {}
  bool operator()(point<DT> p, point<DT> q) {
    return make_tuple(half(p), 0) < make_tuple(half(q),
    (p & q));
  }
};

template <typename DT> class line { public: point<DT>
dir, O; /* direction of vector and starting point */
line(point<DT> p, point<DT> q) : dir(q - p), O(p) {};
bool Contains(const point<double>& p) { return fabs(p -
O & dir) < EPS; } /* checks whether the line Contains a
certain point */ template <typename XT> point<XT> At(XT
t) { return point<XT>(dir) * t + O; } /* inserts value
of t in the vector representation, finds the point which
 is O + Dir*t */ double AtInv(const point<double>& p) {
return abs(dir.x) > 0 ? (p - O).x / dir.x : (p - O).y /
dir.y; } /* if the line Contains a point, gives the
value t such that, p = O+Dir*t */ line Perp(point<DT> p)
 { return line(p, p + (-dir.y, dir.x)); } point<DT>
ProjOfPoint(const point<DT>& P) { return O + dir * ((P -
 O) ^ dir) / (~dir); } double DisOfPoint(const point<DT
>& P) { return fabs(dir & (P - O)) / sqrt(~(dir)); }
friend bool Parallel(line& L, line& R) { return fabs(R.
dir & L.dir) < EPS; } friend int Intersects(line& L,
line& R) { return Parallel(L, R) ? R.Contains(L.O) ? -1
: 0 : 1; } friend pair<double, double> IntersectionAt(
line& L, line& R) { double r = double((L.O - R.O) & L.
dir) / (R.dir & L.dir); double l = double((R.O - L.O) &
R.dir) / (L.dir & R.dir); return {l, r}; } friend pair<
int, point<double>> IntersectionPoint( line L, line R,
```

```
int _L = 0, int _R = 0) { /* _L and _R can be 0 to 3, 0 is a normal line, 3 is a segment, 1 and 2 are rays ( considered bitwise) */ int ok = Intersects(L, R); if (ok == 0) return {0, {0, 0}}; if (ok == 1) { auto [l, r] = IntersectionAt(L, R); if (l < (0 - EPS) and _L & 2) return {0, {0, 0}}; if (l > (1 + EPS) and _L & 1) return {0, {0, 0}}; if (r < (0 - EPS) and _R & 2) return {0, {0, 0}}; if (r > (1 + EPS) and _R & 1) return {0, {0, 0}}; return {1, L.At(l)}; } return {-1, {0, 0}}; /* they are the same line */ } };

template <typename DT> class circle { public: point<DT> O; DT R; circle(const point<DT>& O = {0, 0}, DT R = 0) : O(O), R(R) {} /* the next two make sense only on circle <double> */ circle(const point<DT>& A, const point<DT>& B, const point<DT>& C) { point<DT> X = (A + B) / 2, Y = (B + C) / 2, d1 = Perp(A - B), d2 = Perp(B - C); O = IntersectionPoint(line(X, d1), line(Y, d2)).second; R = sqrt(~(O - A)); } circle(const point<DT>& A, const point<DT>& B, DT R) { point<DT> X = (A + B) / 2, d = Perp(A - B); d = d * (R / sqrt(~(d))); O = X + d; R = sqrt(~(O - A)); } double SectorArea(double ang) { /* Area of a sector of cicle */ return ang * R * R * .5; } double SectorArea(const point<DT>& a, const point<DT>& b) { return SectorArea(Angle(a - O, b - O)); } double ChordArea(const point<DT>& a, const point<DT>& b) { /* Area between sector and its chord */ return SectorArea(a, b) - 0.5 * TriArea(O, a, b); } int Contains(const point<DT>& p) { /* 0 for outside, 1 for inside, -1 for on the circle */ DT d = DisSq(O, p); return d > R * R ? 0 : (d == R * R ? -1 : 1); } friend tuple<int, point<DT>, point<DT>> IntersectionPoint(const circle& a, const circle& b) { if (a.R == b.R and a.O == b.O) return {-1, {0, 0}, {0, 0}}; double d = sqrt(DisSq(a.O, b.O)); if (d > a.R + b.R or d < fabs(a.R - b.R)) return {0, {0, 0}, {0, 0}}; double z = (sq(a.R) + sq(d) - sq(b.R)) / (2 * d); double y = sqrt(sq(a.R) - sq(z)); point<DT> O = b.O - a.O, h = Perp(O) * (y / sqrt(~O)); O = a.O + O * (z / sqrt(~O)); return make_tuple(1 + (~(h) > EPS), O - h, O + h); } friend tuple<int, point<DT>, point<DT>> IntersectionPoint(const circle& C, line<DT> L) { point<DT> P = L.ProjOfPoint(C.O); double D = DisSq(C.O, P); if (D > C.R * C.R) return {0, {0, 0}, {0, 0}}; double x = sqrt(C.R * C.R - D); point<DT> h = L.dir * (x / sqrt(~L.dir)); return {1 + (x > EPS), P - h, P + h}; } double SegmentedArea(point<DT>& a, point<DT>& b) { /* signed area of the intersection between the circle and triangle OAB */ double ans = SectorArea(a, b); line<DT> L(a, b); auto [cnt, p1, p2] = IntersectionPoint(*this, L); if (cnt < 2) return ans; double t1 = L.AtInv(p1), t2 = L.AtInv(p2); if (t2 < 0 or t1 > 1) return ans; if (t1 < 0) p1 = a; if (t2 > 1) p2 = b; return ans - ChordArea(p1, p2); } };

////////////////////////////////////////
namespace polygon_algo {
template <typename DT>
polygon<DT> ConvexHull(polygon<DT> &PT) {
  sort(PT.begin(), PT.end());
  int m = 0, n = PT.size();
  polygon<DT> hull(n + n + 2);
  for (int i = 0; i < n; i++) {
    for (; m > 1 && orientation (hull[m - 2], hull[m - 1], PT[i]) < 0; m--);
    hull[m++] = PT[i];
  }
  for (int i = n - 2, k = m; i >= 0; i--) {
    for (; m > k && orientation (hull[m - 2], hull[m - 1], PT[i]) < 0; m--);
    hull[m++] = PT[i];
  }
  if (n > 1) m--;
  hull.resize(m);
  return hull;
}

template <typename DT> double MinimumBoundingBox(polygon<DT> P) { auto p = ConvexHull(P); int n = p.size(); double area = 1e20 + 5; for (int i = 0, l = 1, r = 1, u = 1; i < n; i++) { point<DT> edge = (p[(i + 1) % n] - p[i]) / sqrt(DisSq(p[i], p[(i + 1) % n])); for (; (edge ^ p[r % n] - p[i]) < (edge ^ p[(r + 1) % n] - p[i]); r++); for (; u < r || (edge & p[u % n] - p[i]) < (edge & p[(u + 1) % n] - p[i]); u++); for (; l < u || (edge ^ p[l % n] - p[i]) > (edge ^ p[(l + 1) % n] - p[i]); l++); double w = (edge ^ p[r % n] - p[i]) - (edge ^ p[l % n] - p[i]); double h = UTriArea(p[u % n], p[i], p[(i + 1) % n]) / sqrt(DisSq(p[i], p[(i + 1) % n])); area = min(area, w * h); } if (area > 1e19) area = 0; return area; }

template <typename DT> DT FarthestPairOfPoints(polygon<DT> p) { p = ConvexHull(p); int n = p.size(); DT ans = -1e9; for (int i = 0, j = 1; i < n; i++) { for (; UTriArea(p[i], p[(i + 1) % n], p[(j + 1) % n]) > UTriArea(p[i], p[(i + 1) % n], p[j]); j = (j + 1) % n); ans = max(ans, DisSq(p[i], p[j])); ans = max(ans, DisSq(p[(i + 1) % n], p[j])); } return ans; /* will return square of the answer. */ }

////////////////////////////////////////
// -1 if strictly inside, 0 if on the polygon, 1 if strictly outside
// it must be strictly convex, otherwise make it strictly convex first
int is_point_in_convex(vector<PT> &p, const PT& x) { // O(log n)
    int n = p.size(); assert(n >= 3);
    int a = orientation(p[0], p[1], x), b = orientation(p[0], p[n - 1], x);
    if (a < 0 || b > 0) return 1;
    int l = 1, r = n - 1;
    while (l + 1 < r) {
        int mid = l + r >> 1;
        if (orientation(p[0], p[mid], x) >= 0) l = mid;
        else r = mid;
    }
    int k = orientation(p[l], p[r], x);
    if (k <= 0) return -k;
    if (l == 1 && a == 0) return 0;
    if (r == n - 1 && b == 0) return 0;
    return -1;
}

// -1 --> outside, 0 --> boundary, 1 --> inside
int pointInPolygon(vector<point> &pol, point z) { int n = pol.size(), winding = 0; for(int i = 0; i < n; i++) { point p1 = pol[i], p2 = pol[(i + 1) % n]; if(isPointOnSegment(p1, p2, z)) return 0; int ori = sign(cross(p1, p2, z)); if(p1.y <= z.y && p2.y > z.y && ori > 0) winding++; if(p1.y > z.y && p2.y <= z.y && ori < 0) winding--; } return winding != 0 ? 1 : -1; }

}  /* namespace polygon_algo */

/*  CLOSEST PAIR OF POINTS */
template <typename DT>
Dis(point<DT> a, point<DT> b) { return ~(a - b); }
```

```
template <typename DT> DT Closest_Distance(vector<point<
DT>>& v) { int n = v.size(); sort(v.begin(), v.end());
auto cmp = [](point<DT> a, point<DT> b) { return (a.y <
b.y || (a.y == b.y && a.x < b.x)); }; set<point<DT>,
decltype(cmp)> s(cmp); DT best = 1e18; int j = 0; for (
int i = 0; i < n; i++) { while (sq(v[i].x - v[j].x) >=
best) { s.erase(v[j]); j = (j + 1) % n; } DT d = best;
auto it1 = s.lower_bound(point<DT>(v[i].x, v[i].y - d));
 auto it2 = s.upper_bound(point<DT>(v[i].x, v[i].y + d))
; for (auto it = it1; it != it2; it++) best = min(best,
Dis(v[i], *it)); s.insert(v[i]); } return best; }
```

7.2 3D

```
template <typename DT> class Point { public: DT x, y, z;
 Point() {}; Point(DT x, DT y, DT z) : x(x), y(y), z(z)
{} Point operator&(const Point& rhs) const { /* cross
product */ return Point(y * rhs.z - z * rhs.y, z * rhs.x
 - x * rhs.z, x * rhs.y - y * rhs.x); } DT operator^(
const Point& rhs) const { /* dot product */ return x *
rhs.x + y * rhs.y + z * rhs.z; } friend DT DisSq(const
Point& a, const Point& b) { return (a.x - b.x) * (a.x -
b.x) + (a.y - b.y) * (a.y - b.y) + (a.z - b.z) * (a.z -
b.z); } };
```

7.3 MinDisSquares

```
typedef long double ld;
const ld eps = 1e-12;
int cmp(ld x, ld y = 0, ld tol = eps) {
  return (x <= y + tol) ? (x + tol < y) ? -1 : 0 : 1;
}
struct point {
  ld x, y;
  point(ld a, ld b) : x(a), y(b) {}
  point() {}
};
struct square {
  ld x1, x2, y1, y2, a, b, c;
  point edges[4];
  square(ld _a, ld _b, ld _c) {
    a = _a, b = _b, c = _c;
    x1 = a - c * 0.5;
    x2 = a + c * 0.5;
    y1 = b - c * 0.5;
    y2 = b + c * 0.5;
    edges[0] = point(x1, y1);
    edges[1] = point(x2, y1);
    edges[2] = point(x2, y2);
    edges[3] = point(x1, y2);
  }
};
ld min_dist(point& a, point& b) {
  ld x = a.x - b.x, y = a.y - b.y;
  return sqrt(x * x + y * y);
}
bool point_in_box(square s1, point p) {
  if (cmp(s1.x1, p.x) != 1 && cmp(s1.x2, p.x) != -1 &&
  cmp(s1.y1, p.y) != 1 &&
      cmp(s1.y2, p.y) != -1)
    return true;
  return false;
}
bool inside(square& s1, square& s2) {
  for (int i = 0; i < 4; ++i)
    if (point_in_box(s2, s1.edges[i])) return true;
  return false;
}
bool inside_vert(square& s1, square& s2) {
  if ((cmp(s1.y1, s2.y1) != -1 && cmp(s1.y1, s2.y2) !=
  1) ||
      (cmp(s1.y2, s2.y1) != -1 && cmp(s1.y2, s2.y2) != 1)
      )
    return true;
  return false;
}
bool inside_hori(square& s1, square& s2) {
  if ((cmp(s1.x1, s2.x1) != -1 && cmp(s1.x1, s2.x2) !=
  1) ||
      (cmp(s1.x2, s2.x1) != -1 && cmp(s1.x2, s2.x2) != 1)
      )
    return true;
  return false;
}
ld min_dist(square& s1, square& s2) {
  if (inside(s1, s2) || inside(s2, s1)) return 0;
  ld ans = 1e100;
  for (int i = 0; i < 4; ++i)
    for (int j = 0; j < 4; ++j)
      ans = min(ans, min_dist(s1.edges[i], s2.edges[j]));
  if (inside_hori(s1, s2) || inside_hori(s2, s1)) {
    if (cmp(s1.y1, s2.y2) != -1)
      ans = min(ans, s1.y1 - s2.y2);
    else if (cmp(s2.y1, s1.y2) != -1)
      ans = min(ans, s2.y1 - s1.y2);
  }
  if (inside_vert(s1, s2) || inside_vert(s2, s1)) {
    if (cmp(s1.x1, s2.x2) != -1)
      ans = min(ans, s1.x1 - s2.x2);
    else if (cmp(s2.x1, s1.x2) != -1)
      ans = min(ans, s2.x1 - s1.x2);
  }
  return ans;
}
```

7.4 Picks Theorem

```
struct Point {
  long long x, y;
  Point() {}
  Point(long long x, long long y) : x(x), y(y) {}
};  /// twice the area of polygon
long long double_area(Point poly[], int n) {
  long long res = 0;
  for (int i = 0, j = n - 1; i < n; j = i++) {
    res += ((poly[j].x + poly[i].x) * (poly[j].y - poly[
    i].y));
  }
  return abs(res);
}  /// number of lattice points strictly on polygon
border
long long on_border(Point poly[], int n) {
  long long res = 0;
  for (int i = 0, j = n - 1; i < n; j = i++) {
    res += __gcd(abs(poly[i].x - poly[j].x), abs(poly[i
    ].y - poly[j].y));
  }
  return res;
}  /// number of lattice points strictly inside polygon
long long interior(Point poly[], int n) {
  long long res = 2 + double_area(poly, n) - on_border(
  poly, n);
  return res / 2;
}
```

7.5 convex

```
/// minkowski sum of two polygons in O(n)
Polygon minkowskiSum(
    Polygon A, Polygon B) {
  int n = A.size(), m = B.size();
```

```
  rotate(A.begin(), min_element(A.begin(), A.end()), A.
  end());
  rotate(B.begin(), min_element(B.begin(), B.end()), B.
  end());
  A.push_back(A[0]);
  B.push_back(B[0]);
  for (int i = 0; i < n; i++) A[i] = A[i + 1] - A[i];
  for (int i = 0; i < m; i++) B[i] = B[i + 1] - B[i];
  Polygon C(n + m + 1);
  C[0] = A.back() + B.back();
  merge(A.begin(), A.end() - 1, B.begin(), B.end() - 1,
  C.begin() + 1,
        polarComp(Point(0, 0), Point(0, -1)));
  for (int i = 1; i < C.size(); i++) C[i] = C[i] + C[i -
   1];
  C.pop_back();
  return C;
}
// finds the rectangle with minimum area enclosing a
convex polygon and
// the rectangle with minimum perimeter enclosing a
convex polygon
// Tf Ti Same
pair<Tf, Tf> rotatingCalipersBoundingBox(const Polygon&
p) {
  using Linear::distancePointLine;
  int n = p.size();
  int l = 1, r = 1, j = 1;
  Tf area = 1e100;
  Tf perimeter = 1e100;
  for (int i = 0; i < n; i++) {
    Point v = (p[(i + 1) % n] - p[i]) / length(p[(i + 1)
     % n] - p[i]);
    while (dcmp(dot(v, p[r % n] - p[i]) - dot(v, p[(r +
    1) % n] - p[i])) < 0)
      r++;
    while (j < r || dcmp(cross(v, p[j % n] - p[i]) -
                         cross(v, p[(j + 1) % n] - p[i])) <
                          0)
      j++;
    while (l < j ||
           dcmp(dot(v, p[l % n] - p[i]) - dot(v, p[(l +
           1) % n] - p[i])) > 0)
      l++;
    Tf w = dot(v, p[r % n] - p[i]) - dot(v, p[l % n] - p
    [i]);
    Tf h = distancePointLine(p[j % n], Line(p[i], p[(i +
     1) % n]));
    area = min(area, w * h);
    perimeter = min(perimeter, 2 * w + 2 * h);
  }
  return make_pair(area, perimeter);
} // returns the left side of polygon u after cutting it
 by ray a->b
Polygon cutPolygon(Polygon u, Point a, Point b) {
  using Linear::lineLineIntersection;
  using Linear::onSegment;
  Polygon ret;
  int n = u.size();
  for (int i = 0; i < n; i++) {
    Point c = u[i], d = u[(i + 1) % n];
    if (dcmp(cross(b - a, c - a)) >= 0) ret.push_back(c)
    ;
    if (dcmp(cross(b - a, d - c)) != 0) {
      Point t;
      lineLineIntersection(a, b - a, c, d - c, t);
      if (onSegment(t, Segment(c, d))) ret.push_back(t);
    }
  }
  return ret;
} /* // returns true if point p is in or on triangle abc
 */
bool pointInTriangle(Point a, Point b, Point c, Point p)
 {
  return dcmp(cross(b - a, p - a)) >= 0 && dcmp(cross(c
  - b, p - b)) >= 0 &&
         dcmp(cross(a - c, p - c)) >= 0;
} /* // pt must be in ccw order with no three collinear
points // returns inside
     = -1, on = 0, outside = 1 */
int pointInConvexPolygon(const Polygon& pt, Point p) {
  int n = pt.size();
  assert(n >= 3);
  int lo = 1, hi = n - 1;
  while (hi - lo > 1) {
    int mid = (lo + hi) / 2;
    if (dcmp(cross(pt[mid] - pt[0], p - pt[0])) > 0)
      lo = mid;
    else
      hi = mid;
  }
  bool in = pointInTriangle(pt[0], pt[lo], pt[hi], p);
  if (!in) return 1;
  if (dcmp(cross(pt[lo] - pt[lo - 1], p - pt[lo - 1]))
  == 0) return 0;
  if (dcmp(cross(pt[hi] - pt[lo], p - pt[lo])) == 0)
  return 0;
  if (dcmp(cross(pt[hi] - pt[(hi + 1) % n], p - pt[(hi +
   1) % n])) == 0)
    return 0;
  return -1;
} /* // Extreme Point for a direction is the farthest
point in that direction //
     u is the direction for extremeness */
int extremePoint(const Polygon& poly, Point u) {
  int n = (int)poly.size();
  int a = 0, b = n;
  while (b - a > 1) {
    int c = (a + b) / 2;
    if (dcmp(dot(poly[c] - poly[(c + 1) % n], u)) >= 0
    &&
        dcmp(dot(poly[c] - poly[(c - 1 + n) % n], u)) >=
        0) {
      return c;
    }
    bool a_up = dcmp(dot(poly[(a + 1) % n] - poly[a], u)
    ) >= 0;
    bool c_up = dcmp(dot(poly[(c + 1) % n] - poly[c], u)
    ) >= 0;
    bool a_above_c = dcmp(dot(poly[a] - poly[c], u)) >
    0;
    if (a_up && !c_up)
      b = c;
    else if (!a_up && c_up)
      a = c;
    else if (a_up && c_up) {
      if (a_above_c)
        b = c;
      else
        a = c;
    } else {
      if (!a_above_c)
        b = c;
      else
        a = c;
    }
  }
  if (dcmp(dot(poly[a] - poly[(a + 1) % n], u)) > 0 &&
```

```
      dcmp(dot(poly[a] - poly[(a - 1 + n) % n], u)) > 0)
    return a;
  return b % n;
}
/* For a convex polygon p and a line l, returns a list
of segments of p
that touch or intersect line l. the i'th segment is
considered (p[i],
p[(i + 1) modulo |p|]) */
// #1 If a segment is collinear with the line, only that
 is returned
// #2 Else if l goes through i'th point, the i'th
segment is added
// Complexity: O(lg |p|)
vector<int> lineConvexPolyIntersection(const Polygon& p,
 Line l) {
  assert((int)p.size() >= 3);
  assert(l.a != l.b);
  int n = p.size();
  vector<int> ret;
  Point v = l.b - l.a;
  int lf = extremePoint(p, rotate90(v));
  int rt = extremePoint(p, rotate90(v) * Ti(-1));
  int olf = orient(l.a, l.b, p[lf]);
  int ort = orient(l.a, l.b, p[rt]);
  if (!olf || !ort) {
    int idx = (!olf ? lf : rt);
    if (orient(l.a, l.b, p[(idx - 1 + n) % n]) == 0)
      ret.push_back((idx - 1 + n) % n);
    else
      ret.push_back(idx);
    return ret;
  }
  if (olf == ort) return ret;
  for (int i = 0; i < 2; ++i) {
    int lo = i ? rt : lf;
    int hi = i ? lf : rt;
    int olo = i ? ort : olf;
    while (true) {
      int gap = (hi - lo + n) % n;
      if (gap < 2) break;
      int mid = (lo + gap / 2) % n;
      int omid = orient(l.a, l.b, p[mid]);
      if (!omid) {
        lo = mid;
        break;
      }
      if (omid == olo)
        lo = mid;
      else
        hi = mid;
    }
    ret.push_back(lo);
  }
  return ret;
} /* Calculate [ACW, CW] tangent pair from an external
point */
constexpr int CW = -1, ACW = 1;
bool isGood(Point u, Point v, Point Q, int dir) {
  return orient(Q, u, v) != -dir;
}
Point better(Point u, Point v, Point Q, int dir) {
  return orient(Q, u, v) == dir ? u : v;
}
Point pointPolyTangent(const Polygon& pt, Point Q, int
dir, int lo, int hi) {
  while (hi - lo > 1) {
    int mid = (lo + hi) / 2;
    bool pvs = isGood(pt[mid], pt[mid - 1], Q, dir);
    bool nxt = isGood(pt[mid], pt[mid + 1], Q, dir);
    if (pvs && nxt) return pt[mid];
    if (!(pvs || nxt)) {
      Point p1 = pointPolyTangent(pt, Q, dir, mid + 1, hi
      );
      Point p2 = pointPolyTangent(pt, Q, dir, lo, mid -
      1);
      return better(p1, p2, Q, dir);
    }
    if (!pvs) {
      if (orient(Q, pt[mid], pt[lo]) == dir)
        hi = mid - 1;
      else if (better(pt[lo], pt[hi], Q, dir) == pt[lo])
        hi = mid - 1;
      else
        lo = mid + 1;
    }
    if (!nxt) {
      if (orient(Q, pt[mid], pt[lo]) == dir)
        lo = mid + 1;
      else if (better(pt[lo], pt[hi], Q, dir) == pt[lo])
        hi = mid - 1;
      else
        lo = mid + 1;
    }
  }
  Point ret = pt[lo];
  for (int i = lo + 1; i <= hi; i++) ret = better(ret,
  pt[i], Q, dir);
  return ret;
} /* [ACW, CW] Tangent */
pair<Point, Point> pointPolyTangents(const Polygon& pt,
Point Q) {
  int n = pt.size();
  Point acw_tan = pointPolyTangent(pt, Q, ACW, 0, n - 1)
  ;
  Point cw_tan = pointPolyTangent(pt, Q, CW, 0, n - 1);
  return make_pair(acw_tan, cw_tan);
}
```

8 Misc

8.1 All Macros

```
//#pragma GCC optimize("Ofast")
//#pragma GCC optimization ("O3")
//#pragma comment(linker, "/stack:200000000")
//#pragma GCC optimize("unroll-loops")
//#pragma GCC target("sse,sse2,sse3,ssse3,sse4,popcnt,
abm,mmx,avx,tune=native")
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
//find_by_order(k) --> returns iterator to the kth
largest element counting from 0
//order_of_key(val) --> returns the number of items in a
 set that are strictly smaller than our item
os.erase (os.find_by_order (os.order_of_key(v[i]))) //to
 erase i-th element from ordered multiset
template <typename DT>
using ordered_set = tree <DT, null_type, less<DT>,
rb_tree_tag,tree_order_statistics_node_update>;
struct custom_hash { static uint64_t splitmix64 (
uint64_t x) { x += 0x9e3779b97f4a7c15; x = (x ^ (x >>
30)) * 0xbf58476d1ce4e5b9; x = (x ^ (x >> 27)) * 0
x94d049bb133111eb; return x ^ (x >> 31); } size_t
operator () (uint64_t x) const { static const uint64_t
FIXED_RANDOM = chrono::steady_clock :: now ().
time_since_epoch ().count (); return splitmix64 (x +
FIXED_RANDOM); }} Rng;
```

```
typedef gp_hash_table<int, int, custom_hash> gp;

int leap_years(int y) { return y / 4 - y / 100 + y /
400; }bool is_leap(int y) { return y % 400 == 0 || (y %
4 == 0 && y % 100 != 0); }bool __builtin_mul_overflow (
type1 a, type2 b, type3 &res)cin.tie(0)->ios_base::
sync_with_stdio(0);int getWeekday (int day, int month,
int year) { if (month <= 2) { month += 12; year -= 1; }
int f = (day + (13 * (month + 1)) / 5 + year + year / 4
- year / 100 + year / 400) % 7; return f;}
```

8.2 StressTest

```
#!/bin/bash
# Call as sh stress.sh ITERATIONS
g++ brute.cpp -o brute # brute solution
g++ cand.cpp -o cand # cand solution
g++ gen.cpp -o gen # test case gen
> all.txt
for i in $(seq 1 "$1") ; do
    echo "Attempt $i/$1"
    ./gen > in.txt
    echo "Attempt $i/$1" >> all.txt
    cat < in.txt >> all.txt
    ./brute < in.txt > out1.txt
    ./cand < in.txt > out2.txt
    diff -y out1.txt out2.txt > diff.txt
    if [ $? -ne 0 ] ; then
        echo -e "\nTest case:"
        cat in.txt
        echo -e "\nOutputs:"
        cat diff.txt
        break
    fi
done
files=("in.txt" "out1.txt" "out2.txt" "diff.txt" "cand"
"brute" "gen")
for file in "${files[@]}"; do
    rm "$file"
done
```

8.3 gen

```
mt19937 rng(random_device{}());
LL randomInt(LL low, LL high) {
  uniform_int_distribution<LL> dist(low, high);
  return dist(rng);
}
vector<pair<int, int>> generate_random_tree(int n) {
  vector<int> prufer(n - 2);
  for (int i = 0; i < n - 2; ++i) prufer[i] = randomInt
  (1, n);
  vector<int> degree(n + 1, 1);
  for (int x : prufer) degree[x]++;
  priority_queue<int, vector<int>, greater<int>> pq;
  for (int i = 1; i <= n; ++i)
    if (degree[i] == 1) pq.push(i);
  vector<pair<int, int>> edges;
  for (int x : prufer) {
    int leaf = pq.top();
    pq.pop();
    edges.emplace_back(leaf, x);
    degree[leaf]--, degree[x]--;
    if (degree[x] == 1) pq.push(x);
  }
  int u = pq.top(); pq.pop();
  int v = pq.top(); pq.pop();
  edges.emplace_back(u, v);
  return edges;
}
vector<int> permutation(int n) {
  vector<int> p(n); iota(p.begin(), p.end(), 1);
  shuffle(p.begin(), p.end(), rng);
  return p;
}
```

8.4 kmp

```
int aut[N][26];
void compute_automaton(string s) {
  s += '#';
  int n = (int)s.size();
  vector<int> pi = prefix_function(s);
  for (int i = 0; i < n; i++) {
    for (int c = 0; c < 26; c++) {
      if (i > 0 && 'a' + c != s[i])
        aut[i][c] = aut[pi[i - 1]][c];
      else
        aut[i][c] = i + ('a' + c == s[i]);
    }
  }
}
```

8.5 sum of floor

```
LL floor_sum(LL n, LL m, LL a, LL b) { LL res = 0; if (a
 < 0) { LL ad = (a - (m - 1)) / m; res += ad * n * (n -
1) / 2; a -= ad * m; } if (b < 0) { LL bd = (b - (m - 1)
) / m; res += bd * n; b -= bd * m; } while (true) { if (
a >= m) { res += (n - 1) * n / 2 * (a / m); a %= m; } if
 (b >= m) { res += n * (b / m); b %= m; } LL ym = (a * n
 + b) / m; if (ym == 0) break; LL xm = ym * m - b; res
+= (n - (xm + a - 1) / a) * ym; n = ym; b = (a - xm % a)
 % a; swap(a, m); } return res; }
```

8.6 sumnpx

```
LL solve(LL n, int x, LL m) { int k = x + 1; vector<LL>
y(k + 1, 0); vector<int> pr; vector<bool> ip(k + 1, true
); y[0] = 0; if (k >= 1) y[1] = 1; for (int i = 2; i <=
k; i++) { if (ip[i]) { pr.push_back(i); y[i] = bigmod(i,
 x, m); } for (int p : pr) { if (i * p > k) break; ip[i
* p] = false; y[i * p] = (y[i] * y[p]) % m; if (i % p ==
 0) break; } } for (int i = 1; i <= k; i++) { y[i] = (y[
i - 1] + y[i]) % m; } if (n <= k) return y[n]; vector<LL
> f(k + 1, 1), iF(k + 1, 1); for (int i = 1; i <= k; i
++) { f[i] = (f[i - 1] * i) % m; } iF[k] = bigmod(f[k],
m - 2, m); for (int i = k - 1; i >= 0; i--) { iF[i] = (
iF[i + 1] * (i + 1)) % m; } vector<LL> L(k + 1), R(k +
1); LL nm = n % m; L[0] = nm; for (int i = 1; i <= k; i
++) { L[i] = (L[i - 1] * (nm - i + m) % m) % m; } R[k] =
 (nm - k + m) % m; for (int i = k - 1; i >= 0; i--) { R[
i] = (R[i + 1] * (nm - i + m) % m) % m; } LL res = 0;
for (int i = 0; i <= k; i++) { LL nu = 1; if (i > 0) nu
= (nu * L[i - 1]) % m; if (i < k) nu = (nu * R[i + 1]) %
 m; LL de = (iF[i] * iF[k - i]) % m; if ((k - i) % 2 ==
1) { de = (m - de) % m; } LL t = (y[i] * nu) % m; t = (t
 * de) % m; res = (res + t) % m; } return res; }
```

9 Equations and Formulas

9.1 Core Binomial Identities

Hockey Stick Identity : $\sum_{k=r}^{n}\binom{k}{r}=\binom{n+1}{r+1}$

Vandermonde Identity : $\sum_{k=0}^{r}\binom{n}{k}\binom{m}{r-k}=\binom{n+m}{r}$

General Convolution : $\sum_{k}\binom{r}{k}\binom{s}{n-k}=\binom{r+s}{n}$

Absorption Identity : $k\binom{n}{k}=n\binom{n-1}{k-1}$

Alternating Sum : $\sum_{k=0}^{r}(-1)^k\binom{n}{k}=(-1)^r\binom{n-1}{r}$

Binomial Square Identity : $\sum_{k=0}^{n}\binom{n}{k}^2=\binom{2n}{n}$

Weighted Binomial Sums : $\sum_{k=0}^{n}k\binom{n}{k}=n2^{n-1}$,

$\sum_{k=0}^{n}k^2\binom{n}{k}=n(n+1)2^{n-2}$

9.2 Stars and Bars

Number of non-negative solutions to $x_1+x_2+\cdots+x_k=n$ is $\binom{n+k-1}{k-1}$

With $x_i\geq 1$, it is $\binom{n-1}{k-1}$

9.3 Catalan Numbers

$C_n=\frac{1}{n+1}\binom{2n}{n}$ $C_0=1, C_1=1$ and $C_n=\sum_{k=0}^{n-1}C_kC_{n-1-k}$

The number of ways to completely parenthesize n+1 factors.

The number of triangulations of a convex polygon with n+2 sides (i.e. the number of partitions of polygon into disjoint triangles by using the diagonals).

The number of ways to connect the $2n$ points on a circle to form n disjoint i.e. non-intersecting chords.

The number of rooted full binary trees with n+1 leaves (vertices are not numbered). A rooted binary tree is full if every vertex has either two children or no children.

Number of permutations of $1,\ldots,n$ that avoid the pattern 123 (or any of the other patterns of length 3); that is, the number of permutations with no three-term increasing sub-sequence. For $n=3$, these permutations are 132, 213, 231, 312 and 321.

9.4 Stirling Numbers First Kind

The Stirling numbers of the first kind count permutations according to their number of cycles (counting fixed points as cycles of length one).

$S(n,k)$ counts the number of permutations of n elements with k disjoint cycles.

$S(n,k)=(n-1)\cdot S(n-1,k)+S(n-1,k-1)$, *where*, $S(0,0)=1, S(n,0)=S(0,n)=0$ $\sum_{k=0}^{n}S(n,k)=n!$

The unsigned Stirling numbers may also be defined algebraically, as the coefficient of the rising factorial:

$$x^{\bar{n}}=x(x+1)...(x+n-1)=\sum_{k=0}^{n}S(n,k)x^k$$

Lets $[n,k]$ be the stirling number of the first kind, then

$$\left[{n \atop k}\right]=\sum_{0\leq i_1<i_2<i_k<n}i_1i_2....i_k.$$

9.5 Stirling Numbers Second Kind

Stirling number of the second kind is the number of ways to partition a set of n objects into k non-empty subsets.

$S(n,k)=k\cdot S(n-1,k)+S(n-1,k-1)$, *where* $S(0,0)=1, S(n,0)=S(0,n)=0$ $S(n,2)=2^{n-1}-1$ $S(n,k)\cdot k!=$ number of ways to color n nodes using colors from 1 to k such that each color is used at least once.

An r-associated Stirling number of the second kind is the number of ways to partition a set of n objects into k subsets, with each subset containing at least r elements. It is denoted by $S_r(n,k)$ and obeys the recurrence relation.

$S_r(n+1,k)=kS_r(n,k)+\binom{n}{r-1}S_r(n-r+1,k-1)$

Used in partition DP. $S(n,k)=\frac{1}{k!}\sum_{i=0}^{k}(-1)^i\binom{k}{i}(k-i)^n$

Denote the n objects to partition by the integers $1,2,\ldots,n$.

Define the reduced Stirling numbers of the second kind, denoted $S^d(n,k)$, to be the number of ways to partition the integers $1,2,\ldots,n$ into k nonempty subsets such that all elements in each subset have pairwise distance at least d. That is, for any integers i and j in a given subset, it is required that $|i-j|\geq d$. It has been shown that these numbers satisfy, $S^d(n,k)=S(n-d+1,k-d+1), n\geq k\geq d$

9.6 Other Combinatorial Identities

$\binom{n}{k}=\frac{n}{k}\binom{n-1}{k-1}$

$\sum_{i=0}^{k}\binom{n+i}{i}=\sum_{i=0}^{k}\binom{n+i}{n}=\binom{n+k+1}{k}$

$n,r\in N, n>r, \sum_{i=r}^{n}\binom{i}{r}=\binom{n+1}{r+1}$

If $P(n)=\sum_{k=0}^{n}\binom{n}{k}\cdot Q(k)$, then,

$$Q(n)=\sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}\cdot P(k)$$

If $P(n)=\sum_{k=0}^{n}(-1)^k\binom{n}{k}\cdot Q(k)$, then,

$$Q(n)=\sum_{k=0}^{n}(-1)^k\binom{n}{k}\cdot P(k)$$

Inclusion-Exclusion Template : Number of valid configurations: $\sum_{k=0}^{n}(-1)^k\binom{n}{k}F(k)$

Advanced Summation Identity : $\sum_{k=r}^{n}\binom{k}{r}\binom{n-k}{m}=\binom{n+1}{r+m+1}$

Useful Summation Transform : $\sum_{k=0}^{n}\binom{n}{k}x^ky^{n-k}=(x+y)^n$

Subset Convolution Identity : $\sum_{k=0}^{n}\binom{n}{k}a^kb^{n-k}=(a+b)^n$

9.7 Prime Mod Tricks

Fermat Inverse : For prime p: $a^{-1} \equiv a^{p-2} \pmod{p}$

Lucas Theorem : $\binom{n}{r} \equiv \prod \binom{n_i}{r_i} \pmod{p}$, where n_i, r_i are base-p digits.

9.8 Generating Functions

Binomial Theorem : $(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$

Derivative Trick : $\sum k\binom{n}{k} x^k = nx(1+x)^{n-1}$

Exponential Generating Functions : Useful in counting labeled structures. $\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$

9.9 Tree DP Combinatorics

When merging subtree sizes: $\binom{a+b}{a} = \frac{(a+b)!}{a!b!}$

Used for counting ways to interleave children orders.

9.10 Different Math Formulas

Picks Theorem : $A = i + b/2 - 1$

Derangements : $d(i) = (i-1) \times (d(i-1) + d(i-2))$

Derangements (Explicit) : $D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$

$$\frac{n}{ab} \quad - \quad \left\{\frac{bl n}{a}\right\} \quad - \quad \left\{\frac{al n}{b}\right\} \quad + \quad 1$$

9.11 GCD and LCM

if m is any integer, then $\gcd(a + m \cdot b, b) = \gcd(a, b)$

The gcd is a multiplicative function in the following sense: if a_1 and a_2 are relatively prime, then $\gcd(a_1 \cdot a_2, b) = \gcd(a_1, b) \cdot \gcd(a_2, b)$.

$\gcd(a, \operatorname{lcm}(b, c)) = \operatorname{lcm}(\gcd(a, b), \gcd(a, c))$.

$\operatorname{lcm}(a, \gcd(b, c)) = \gcd(\operatorname{lcm}(a, b), \operatorname{lcm}(a, c))$.

For non-negative integers a and b, where a and b are not both zero, $\gcd(n^a - 1, n^b - 1) = n^{\gcd(a,b)} - 1$

$$\gcd(a, b) = \sum_{k|a \text{ and } k|b} \phi(k)$$

$$\sum_{i=1}^{n} [\gcd(i, n) = k] = \phi\left(\frac{n}{k}\right)$$

$$\sum_{k=1}^{n} \gcd(k, n) = \sum_{d|n} d \cdot \phi\left(\frac{n}{d}\right)$$

$$\sum_{k=1}^{n} x^{\gcd(k,n)} = \sum_{d|n} x^d \cdot \phi\left(\frac{n}{d}\right)$$

$$\sum_{k=1}^{n} \frac{1}{\gcd(k, n)} = \sum_{d|n} \frac{1}{d} \cdot \phi\left(\frac{n}{d}\right) = \frac{1}{n} \sum_{d|n} d \cdot \phi(d)$$

$$\sum_{k=1}^{n} \frac{k}{\gcd(k, n)} = \frac{n}{2} \cdot \sum_{d|n} \frac{1}{d} \cdot \phi\left(\frac{n}{d}\right) = \frac{n}{2} \cdot \frac{1}{n} \cdot \sum_{d|n} d \cdot \phi(d)$$

$$\sum_{k=1}^{n} \frac{n}{\gcd(k, n)} = 2 * \sum_{k=1}^{n} \frac{k}{\gcd(k, n)} - 1, \text{ for } n > 1$$

$$\sum_{i=1}^{n} \sum_{j=1}^{n} [\gcd(i, j) = 1] = \sum_{d=1}^{n} \mu(d) \lfloor \frac{n}{d} \rfloor^2$$

$$\sum_{i=1}^{n} \sum_{j=1}^{n} \gcd(i, j) = \sum_{d=1}^{n} \phi(d) \lfloor \frac{n}{d} \rfloor^2$$

$$\sum_{i=1}^{n} \sum_{j=1}^{n} i \cdot j [\gcd(i, j) = 1] = \sum_{i=1}^{n} \phi(i) i^2$$

$$F(n) = \sum_{i=1}^{n} \sum_{j=1}^{n} \operatorname{lcm}(i, j) = \sum_{l=1}^{n} \left(\frac{\left(1 + \lfloor \frac{n}{l} \rfloor\right) \left(\lfloor \frac{n}{l} \rfloor\right)}{2} \right)^2 \sum_{d|l} \mu(d) l d$$

9.12 Common Optimization Patterns

- Convert double binomial sum → Vandermonde.
- Convert $k\binom{n}{k}$ → Absorption identity.
- Convert alternating prefix → closed binomial.
- Use binomial inversion to undo convolution.
- Use Lucas when n is large.
- Use generating function derivative for weighted sums.

10 Competition Checklist

10.1 Avoiding WA

1. Clear all global arrays.
2. Check `<=, >=, <, >` operators.
3. Verify input reading; check for premature `break` or `return`.
4. Check for division by zero.
5. Verify array, vector, and container sizes.
6. Check for infinite loops in `while` structures.
7. Ensure DP states are being saved.
8. Check for use-after-delete errors.
9. Re-read the problem to ensure correct understanding.
10. Review all problem constraints.
11. Test small cases: $0, 1$, etc.
12. Check for integer overflow (`long long`).
13. Initialize all variables properly.
14. Verify correct application of formulas.

10.2 Stress Testing

1. Verify the stress test script is functioning.
2. Test against multiple cases.
3. Test with both large and very small inputs.
4. Verify the correctness of the brute force code.

10.3 Idea Generation

1. Try thinking in reverse.
2. Re-read the problem and constraints.
3. Expand from small manual cases in notebook.
4. Identify invariants.
5. Consider simpler/greedy ideas.
6. Analyze brute force for optimization patterns.
7. Take a deep breath or a short break.